

Project Report
on
**AI BASED SMART LAND DOCUMENT
ANALYSIS & REGISTRY ASSISTANT
SYSTEM**
at
Institute of Technology



Internal Guide:
Prof. Hiten M Sadani

Prepared By:
Mr. Parva Patel (24172022023)
Mr. Prit Patel (24172022034)

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INSTITUTE OF TECHNOLOGY



18/04/2026

CERTIFICATE

TO WHOM SO EVER IT MAY CONCERN

This is to certify that Mr. Patel Parva VipulKumar (24172022023) student of **B.Tech. Semester VI (Information Technology)** has completed his full semester Capstone Project-II work titled **AI BASED SMART LAND DOCUMENT ANALYSIS & REGISTRY ASSISTANT SYSTEM** satisfactorily in partial fulfillment of the requirement of Bachelor of Technology degree of Information Technology of Ganpat University, Ganpat Vidyanagar, Mehsana in the year 2025-2026.

Prof. Hiten M Sadani
College Project Guide

Dr. Devang S. Pandya
Head of the Department
Information Technology

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This project has been a valuable learning experience that allowed us to apply theoretical knowledge to practical problems, enhance our technical skills, and gain a deeper understanding of real-world system development.

With Sincere Regards from,
Mr. Patel Prava [24172022023]
Mr. Patel Prit [24172022034]

ABSTRACT

The real estate and land registry sector in India faces persistent challenges related to document complexity, fraud risks, and lack of transparency in transactions. The AI-Based Smart Land Document Analysis and Registry Assistant System is designed to modernize land registry workflows through intelligent document processing, risk analysis, and user-centric interaction. The system follows a structured pipeline in which users upload land-related documents such as Index-II and 7/12 records through a frontend interface, and the backend processes these files using advanced OCR techniques to extract structured metadata, including owner details, survey numbers, transaction history, and legal attributes. The extracted data is normalized and stored for further analysis and retrieval. An AI-driven analysis layer evaluates the processed content to identify inconsistencies, detect potential fraud patterns, and highlight legal risks by analyzing anomalies such as duplicate ownership claims, mismatched records, and irregular transaction sequences, ultimately generating a structured risk assessment report. The platform also includes a unified chat interface and voice-enabled system that allows real-time interaction, providing users with contextual explanations, legal guidance, and document summaries to improve accessibility and decision-making. Legal automation modules generate documents such as sale deeds and affidavits based on extracted data and predefined templates, reducing manual effort and ensuring consistency. A smart mapping component enables visualization of land parcels using survey numbers, supporting spatial understanding of property boundaries and ownership distribution. Additionally, financial tools such as stamp duty calculation and loan eligibility analysis provide accurate financial insights based on transaction data and user inputs. The overall system is built on a modular architecture to ensure scalability, maintainability, and efficient data flow, thereby enhancing transparency, reducing fraud risks, and improving the efficiency of land registry process.

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1. INTRODUCTION

This chapter provides an overview of the problem domain, objectives, and purpose of the proposed system. It highlights the challenges faced in the current land registry process and explains how the proposed solution aims to address these issues using modern technologies. The chapter also gives a high-level understanding of the system and its key functionalities.

1.1 Problem Statement

The process of registering land and verifying property documents in regions like Gujarat is slow, fragmented, and difficult for common people to understand, as it involves complex legal procedures, technical terminology, outdated formats, and inconsistent record-keeping systems; important documents such as Index-II, 7/12 extracts, and property cards are not only hard to interpret but also require cross-verification across multiple offices or portals, increasing dependency on middlemen and legal experts, while this manual and semi-digital approach causes delays, long queues, and frequent human errors, with no centralized system to automatically verify ownership history, detect duplicate claims, or identify fraudulent documents in real time, exposing buyers to risks like forged records, unclear titles, and legal disputes; additionally, the lack of user-friendly platforms and proper support in local languages like Gujarati makes it even more difficult for rural and less-educated users to access and understand information, forcing people to rely on agents and spend extra money, which ultimately leads to financial losses, fraud, and prolonged court cases, highlighting the urgent need for a modern, transparent, and intelligent system that simplifies land record verification and reduces dependency on intermediaries.

1.2 Objective

The main goal of this project is to develop an AI-Based Smart Land Document Analysis and Registry Assistant System that simplifies and secures the land registration process for users by reducing complexity and manual effort. The system will leverage high-speed AI-powered document scanning to read and analyze land records, automatically extracting critical information such as ownership details, survey numbers, transaction history, and legal status. It will also incorporate intelligent fraud detection mechanisms to identify fake documents, detect duplicate ownership claims, and verify the authenticity of records, thereby protecting buyers from potential scams and financial losses. A key objective is to provide a user-friendly interaction experience through chat and voice-based assistance in regional languages like Gujarati, along with Hindi and English, ensuring accessibility for users from diverse backgrounds, especially in rural areas. Furthermore, the system aims to automate the generation of essential legal documents such as Sale Deeds, reducing dependency on legal experts, and integrate mapping functionality that uses survey numbers to visually display land boundaries on an interactive map, helping users better understand property details. Overall, the project

focuses on enhancing transparency, accuracy, accessibility, and efficiency in the land registration ecosystem.

1.3 Purpose

The purpose of this project is to develop a modern, integrated platform that makes land transactions clear, secure, and easy to understand by combining all essential processes—ranging from document verification to legal draft generation—into a single system. It aims to eliminate the complexity and manual effort involved in checking property background details by using intelligent automation to quickly analyze records, identify hidden issues such as legal disputes, unpaid debts, or ownership inconsistencies, and present the information in a simple and understandable format. By providing accurate insights and reducing reliance on intermediaries, the platform empowers users to make informed investment decisions with confidence. Additionally, by actively detecting fraud and ensuring document authenticity, the system significantly enhances the safety and transparency of land dealings, making property transactions more reliable, efficient, and accessible for all users, including those with limited legal or technical knowledge.

1.4 Overview

The AI-Based Smart Land Document Analysis & Registry Assistant System is an all-in-one digital platform designed to streamline and simplify land-related services by integrating document analysis, verification, and legal assistance into a single system. It enables users to upload land documents and receive instant summaries with key details such as ownership, transaction history, and legal status, while also providing features to verify land ownership and check official market rates (Jantri). The system uses advanced AI to break down complex legal terminology into simple, understandable language and performs risk assessment to identify suspicious or potentially fraudulent documents. In addition, it offers a user-friendly interface with both voice and text interaction in multiple languages, making it accessible to a wider audience. The platform also assists in generating essential legal drafts such as sale agreements and provides interactive map-based visualization of land locations using survey numbers, ensuring users have a complete, clear, and secure understanding of property details before making decisions.

2. PRJOCET SCOPE

This chapter provides an overview of the Ai Based Smart Land Document Analysis & Registry Assistant System by defining its scope, requirements, and research background. It explains the functional and non-functional aspects of the system, including user and admin features. The chapter also analyzes existing land record platforms to identify gaps and justify the need for an intelligent solution. Overall, it lays the foundation for designing an efficient and user-friendly system.

2.1 Project scope

This project focuses on the design and development of an intelligent land record management and legal advisory system that integrates automation, document analysis, and decision-support capabilities. The AI-based smart land document analysis and registry assistant system addresses limitations in existing land registry platforms by enabling accurate interpretation of legal records, fraud detection, and user-centric guidance. The scope includes analysis of existing systems such as AnyRoR (Gujarat), Digital India Land Records (DILRMP), and other legal technology platforms to identify gaps in usability, personalization, and document understanding. Current platforms primarily function as data repositories and lack intelligent interpretation, multilingual accessibility, and advisory support, which this project overcomes through AI-driven features like 7/12 record interpretation, automated legal guidance, and dynamic government form retrieval to enhance usability and decision-making.

The project scope also includes a detailed literature review and domain analysis covering Optical Character Recognition (OCR) for regional languages, semantic search in legal documents, and conversational AI interfaces. It examines workflows in the land revenue and registration sector, including 7/12 Utara records, Index-II documents, Jantri rate calculations, RERA processes, and ownership transfer protocols, while integrating factors such as tenure classification and encumbrances for accurate risk assessment. User-centric research identifies key issues like manual document analysis, lack of legal clarity, and fragmented information access, leading to features such as AI-generated summaries, guided workflows, and automated legal drafting. The technical scope involves integration of OCR engines, large language models (Google Gemini), vector databases, and a Retrieval-Augmented Generation (RAG) framework, along with a modular MERN-based architecture supporting document processing, fraud detection, mapping visualization, financial analysis, and an admin module for regulatory updates.

2.2 Functional Requirements

Functional requirements define the core features and operations that a system must perform to meet user needs. They specify how the system behaves, including inputs, processes, and outputs for different user roles. These requirements ensure that all essential functionalities, such as data handling, user interaction, and system responses, are clearly defined and properly implemented.

2.2.1 Admin Panel Requirements

The Admin Panel is designed to provide secure and efficient management of the system. It includes an authentication module secured using JSON Web Token (JWT) to ensure that only authorized users can access the system. Admins can upload and manage important documents such as rule books, Jantri PDFs, and Real Estate Regulatory Authority (RERA) related files. It also provides a review system with an editor to correct text extracted using Optical Character Recognition (OCR) technology. Additionally, the system maintains proper data synchronization by allowing deletion or archiving of outdated rules when new rules are added.

2.2.2 User Panel Requirements

The User Panel is designed to provide a simple and user-friendly interface for interacting with the system. It allows users to upload documents in PDF or image format with a maximum size of 5 MB. The panel includes an AI chat interface that supports real-time communication in Hinglish and Gujarati languages. Users can also download generated forms or guides directly from the chat. Additionally, it ensures privacy by using session-based storage, where data is only retained during the active chat session.

2.3 Non-Functional Requirements

The Non-Functional Requirements define the quality and performance aspects of the system. The system ensures availability by using loading animations to manage cloud cold-start delays of 30–50 seconds. For security, Application Programming Interface (API) keys are stored safely using server-side environment variables (.env), and user documents are encrypted before being stored in MongoDB (a NoSQL database). The interface is designed with high usability, making it simple and easy to use even for users with low technical knowledge. To maintain scalability, request queuing is implemented to handle limits of the Google Gemini API (Artificial Intelligence service by Google). Additionally, reliability is ensured through an exponential back off and retry mechanism to manage temporary failures efficiently.

2.4 Literature Review

Introduction Land records management has seen significant technological investment globally, yet a wide "Digital Divide" persists. While governments have successfully moved records from dusty archives to digital databases, the interface for the end-user remains archaic. This literature review analyzes current market implementations, specifically official state portals, and highlights critical gaps in user assistance and intelligent analysis that have hindered the accessibility of land records for common citizens.

Analysis of Market Portals: The Case of AnyROR (Gujarat) In the current market, official platforms like AnyROR (Gujarat) and E-Dhara serve as the primary sources for 7/12, 8A, and Index-II documents. While these systems are efficient at storing and retrieving data, they operate as purely "Static Portals." A significant problem identified in these market systems is the total lack of user guidance. The AnyROR site, for instance, provides no real-time support, meaning a user must already possess technical and legal knowledge to navigate the complex menus, find their specific record, and interpret the legal terminology used within it.

The "Assistance Gap" and Missing Interaction One of the most glaring limitations of existing market projects is the absence of a Unified Chat Interface. When a citizen downloads a land record from a government site, they are often left with questions regarding "Boj" (encumbrances), "Jantri" rates, or legal remarks. Current systems do not provide a way to ask these questions or get instant clarifications. Our research indicates that the lack of an integrated chat or assistant feature forces users to rely on expensive intermediaries or local agents, which increases the cost of transactions and the risk of being misled.

Static Data vs. Proactive Intelligent Analysis Existing market solutions treat land records as "images" or "dead data." They do not perform any proactive analysis for fraud or risk. For example, a government portal will show an Index-II document even if it contains suspicious patterns or duplicate claims, as long as it exists in the database. There is no system in the current market that uses AI to "read" the document semantically, check for fraudulent indicators, or automatically generate legal sale deeds based on the analyzed data.

3. FEASIBILITY ANALYSIS

This chapter evaluates the feasibility of the proposed AI Based Smart Land Document Analysis & Registry Assistant System from multiple perspectives, including technical, time, operational, implementation, and economic aspects. It examines whether the system can be developed with available technologies, completed within the given timeline, and effectively used in real-world scenarios. The analysis also highlights cost considerations and overall practicality, ensuring the project is realistic and achievable.

3.1 Technical Feasibility

Technical feasibility evaluates whether the required technology and resources are available for system development, and the AI-Based Smart Land Document Analysis & Registry Assistant System is clearly feasible using a modern and proven technology stack. The frontend can be built using React.js along with CSS or Tailwind to create a responsive and user-friendly interface, while the backend can be developed using Node.js and Express.js to efficiently handle API requests and implement business logic. MongoDB can be used as the database to store user profiles, land records, and document-related metadata in a flexible and scalable manner. For core AI functionality, advanced Optical Character Recognition (OCR) tools and Large Language Models (LLMs) can be integrated to process and analyze land documents, enabling extraction of key information, simplification of complex legal language, and detection of inconsistencies or fraudulent data. The system can be deployed on reliable cloud platforms such as Amazon Web Services, Google Cloud, or Microsoft Azure to ensure high availability, secure data storage, and scalable computing resources. Since all these technologies are mature, widely used, and well-supported in the industry, the project is technically practical and achievable without major limitations.

3.2 Time Schedule Feasibility

Time feasibility determines whether the project can be completed within the available time frame, and for the AI-Based Smart Land Document Analysis & Registry Assistant System, the timeline is realistic and achievable with proper planning. The development process is divided into structured phases, including requirement gathering, system design (UI/UX), frontend and backend development, AI model integration, testing, and final deployment, ensuring a clear and organized workflow. The entire system can be developed and thoroughly tested within approximately 20–24 weeks, which aligns well with a typical academic semester schedule. By adopting an Agile development model, the team can work in iterative cycles, starting with the basic user and admin functionalities, followed by integration of AI-based document analysis features, and finally refining the application through testing and improvements. This phased and iterative approach ensures continuous progress, better risk management, and timely completion of the project within the defined duration.

3.3 Operational Feasibility

Operational feasibility evaluates whether the system will function effectively in a real-world environment and satisfy user requirements, and the AI-Based Smart Land Document Analysis & Registry Assistant System is designed with strong practical usability in mind. The system streamlines the traditionally complex and time-consuming process of land registration and document verification by enabling users such as property buyers, sellers, and legal consultants to securely upload documents and receive instant AI-based analysis that checks authenticity, identifies missing clauses, and highlights potential risks, while also allowing users to track the status of their applications. On the administrative side, the system provides a dedicated panel for registrars and officials to manage verified records, monitor user activities, and oversee registry operations efficiently. The interface is intentionally designed to be simple and intuitive so that even non-technical users can easily interact with the system, upload documents, and understand the generated reports without external help. By reducing manual paperwork, minimizing human errors, and speeding up processing time, the system offers strong practical value and proves to be operationally feasible in real-world conditions.

3.4 Implementation Feasibility

Implementation feasibility analyzes whether the system can be successfully deployed and effectively used in a real-world environment, and the AI-Based Smart Land Document Analysis & Registry Assistant System is well-suited for practical implementation using modern technologies. The system can be easily deployed on secure cloud infrastructure, ensuring reliability, scalability, and data protection, with strict role-based access control (RBAC) mechanisms in place to safeguard sensitive land and user data. The platform is designed with separate interfaces to enhance usability, where general users can interact seamlessly through a web application to upload documents and access services, while administrators can manage operations, monitor activities, and control system workflows through a comprehensive dashboard. Since the system automates complex legal analysis and presents outputs in a simple and understandable manner, it requires minimal training for users, making adoption easier even for non-technical individuals. For effective rollout, the implementation can follow a strategic approach by initially launching a pilot program within a limited region, such as a specific municipality, and then gradually scaling the system to support a larger user base based on feedback and performance, ensuring a smooth and successful deployment.

3.5 Economic Feasibility

Economic feasibility evaluates the financial viability of the project, and the AI-Based Smart Land Document Analysis & Registry Assistant System is cost-effective due to the use of open-source technologies like React, Node.js, and MongoDB. The main costs involve API usage, domain setup, and optional hosting.

4. SOFTWARE AND HARDWARE REQUIREMENTS

This chapter defines the software and hardware requirements needed to develop and run the AI Based Smart Land Document Analysis & Registry Assistant System. It outlines the technologies used for frontend, backend, AI integration, and data storage, along with the necessary system specifications for developers and end-users. The goal is to ensure smooth development, efficient performance, and accessibility across different devices.

4.1 Software Requirements

Software requirements define the essential tools, technologies, and platforms needed to develop and run the system. They ensure smooth functioning, performance, and scalability of the application. These requirements include frontend, backend, database, and AI technologies required for system implementation.

4.1.1 Frontend Technology (Client-Side)

The frontend of the system is developed using React.js with Vite, which enables fast development, efficient hot module replacement, and optimized production builds. For styling, Tailwind CSS is used to create a responsive, modern, and consistent user interface for both the Admin and User panels. The UI components are built using Shadcn UI and Headless UI, providing reusable, accessible, and customizable elements. Icons are implemented using Lucide React for a clean and scalable visual design. State management is handled through the React Context API, which manages authentication, application state, and chat session data. Additionally, Axios is used to facilitate seamless communication between the frontend and backend through REST APIs.

4.1.2 Backend Technology (Server-Side)

The backend of the system is built using Node.js, which provides a scalable and efficient environment for server-side execution. The application uses Express.js as the framework to develop RESTful APIs and handle middleware operations effectively. For authentication, JSON Web Token is implemented to ensure secure access for both Admin and User roles. File handling is managed using Multer, which processes PDF and image uploads efficiently. Additionally, data validation is performed using Zod or Joi to ensure request data is accurate, secure, and maintains overall data integrity.

4.1.3 Artificial Intelligence & RAG Stack

The system integrates advanced AI technologies to enable intelligent document processing and user interaction. The Large Language Model used is Google Gemini 2.5 Flash, which provides fast response generation, supports large context handling, and ensures cost-effective inference. For semantic understanding, text-embedding-004 is used to convert text into vector representations. Text extraction from scanned land documents is handled by Tesseract.js, which supports both Gujarati and English languages. Additionally, LangChain.js is used as the RAG framework to connect the

language model with the vector database, while managing retrieval logic and prompt engineering efficiently.

4.1.4 Database & Storage

The system uses MongoDB as the primary database to store metadata, user profiles, and system logs. Running MongoDB locally provides full control over data management and eliminates dependency on cloud services during development. For semantic search in the RAG pipeline, MongoDB Vector Search is utilized to store embeddings and perform similarity-based retrieval. Additionally, Cloudinary is used as external file storage to host uploaded PDF and image files, reducing database load and improving overall system performance.

Category	Technology / Tool	Description
Frontend (Client-Side)	React.js + Vite	Used for building fast and efficient user interfaces with optimized performance and hot module replacement
	Tailwind CSS	Provides responsive, modern, and consistent UI design
	Shaden UI / Headless UI	Offers reusable, accessible, and customizable UI components
	Lucid React	Used for scalable and clean icons
	React Context API	Manages global state such as authentication and chat sessions
	Axios	Handles API communication between frontend and backend
Backend (Server-Side)	Node.js	Provides scalable server-side runtime environment
	Express.js	Used to build RESTful APIs and manage middleware
	JSON Web Token (JWT)	Ensures secure authentication and authorization
	Multer	Handles file uploads (PDFs and images)
	Zod / Joi	Performs data validation to maintain data integrity
AI & RAG Stack	Google Gemini 2.5 Flash	Large Language Model for fast and efficient response generation
	text-embedding-004	Converts text into vector embeddings for semantic understanding
	Tesseract.js	Performs OCR for Gujarati and English documents
	LangChain.js	Manages RAG pipeline, retrieval logic, and prompt engineering
Database & Storage	MongoDB	Stores user data, metadata, and system logs
	MongoDB Vector Search	Enables similarity-based search using embeddings
	Cloud nary	Stores uploaded files (PDFs/images) externally

Table 1 Software Requirement

The software requirements used for the development of the proposed system are presented in Table 1. It includes the key technologies and tools utilized across different layers such as frontend, backend, artificial intelligence, and database management. These technologies are selected to ensure efficient system performance, scalability, security, and intelligent data processing. The combination of modern frameworks and AI tools enables the system to deliver accurate, fast, and user.

4.2 Hardware Requirements

Hardware requirements define the physical components needed to develop, deploy, and use the system effectively. They include devices such as computers, servers, and user devices with sufficient processing power, memory, and storage. These requirements ensure smooth performance, fast processing, and reliable system operation.

4.2.1 Developer Machine

The system requires a minimum hardware configuration to ensure smooth development and execution. A processor such as Intel Core i3 7th Generation or AMD Ryzen 3 (or higher) is recommended to handle processing tasks efficiently. At least 8 GB of RAM is required due to memory-intensive operations, including Node.js services and OCR processing. For storage, a 256 GB SSD is preferred to enable faster builds, compilation, and local testing. Additionally, a minimum internet connectivity of 2 Mbps broadband is necessary for API communication and deployment-related tasks.

4.2.2 End-User Device

The system is accessible across multiple devices with minimal hardware requirements. Smartphones with at least 2 GB RAM and a modern web browser such as Google Chrome or Safari can efficiently access the application. For desktop or laptop users, any basic system capable of running a modern web browser is sufficient, as most processing is handled by the backend server rather than the client device.

Category	Component	Specification
Developer Machine	Processor	Intel Core i3 (7th Gen) or AMD Ryzen 3 (or higher)
	RAM	Minimum 8 GB
	Storage	Minimum 256 GB SSD
	Internet	Minimum 2 Mbps broadband
End-User Device	Mobile Device	Smartphone with minimum 2 GB RAM
	Browser (Mobile)	Google Chrome / Safari
	Desktop/Laptop	Any system capable of running a modern web browser
	Browser (Desktop)	Google Chrome or any modern browser

Table 2 Hardware Requirement

5. PROCESS MODEL

This chapter describes the development methodology used for the AI Based Smart Land Document Analysis & Registry Assistant System, specifically the Incremental Model. It explains how the system is built in multiple stages, where each phase adds new functionality and improves the overall system. The chapter outlines all stages of development, from planning to maintenance, ensuring a structured, scalable, and efficient implementation process.

5.1 Incremental Model

The Incremental Model is a process of software development where requirements are divided into multiple standalone modules of the software development cycle. In this model, each module goes through the requirements, design, implementation, and testing phases. Every subsequent release of the module adds function to the previous release. The process continues until the complete system is achieved. Each iteration passes through the requirements, design, coding, and testing phases. For this Land Record RAG Chatbot, the development is divided into increments such as the Core Retrieval Engine, OCR Integration for 7/12 documents, and the Admin Knowledge Management Module. Each subsequent release of the system adds functionality to the previous release until all designed functionality, including legal advisory and automated form retrieval, has been implemented.

5.1.1 Planning Stage

The project planning phase focuses on defining clear objectives, scope, resources, and risk management strategies for successful execution. The primary objective is to develop an AI-driven RAG (Retrieval-Augmented Generation) system for land records such as 7/12 Utara, along with an automated Legal Advisor for property-related queries. The project scope and constraints are defined using the MERN stack integrated with Google Gemini, including limitations on document size, OCR processing, and API usage. Resources and timelines are planned based on local development infrastructure, including system hardware and required software tools. A structured schedule is prepared with key milestones such as vector search implementation, OCR module development, and final system integration. Potential risks, including OCR inaccuracies in regional languages and API rate limits, are identified and addressed through basic risk mitigation strategies.

5.1.2 Analysis Stage

The requirement analysis phase focuses on understanding system needs, validating feasibility, and defining core modules. Detailed requirements are gathered from both users and administrators by analyzing their needs related to 7/12 documents, Jantri rates, RERA guidelines, and land transfer processes. Technical and economic feasibility analysis is conducted to evaluate the effectiveness of Tesseract.js for OCR processing and local MongoDB for managing data and supporting semantic search operations. The major system modules and data requirements are identified, including embeddings

generation, document retrieval, and the Admin Content Management System. Finally, a high-level system architecture is outlined to define the complete data flow, starting from document ingestion to AI-generated responses.

5.1.3 Design Stage

The system design phase focuses on defining the overall architecture, database structure, and workflow logic. The system architecture is designed to establish clear communication between the React frontend, Node.js backend, and Google Gemini API services for seamless data exchange. The database schema is structured using MongoDB, with collections designed for the Global Knowledge Base (laws and regulations), user-uploaded documents, and embeddings data. UI/UX layouts are prepared to ensure a responsive chat interface for users and an efficient admin dashboard for content management. Additionally, document processing workflows are defined, including steps such as image preprocessing, OCR-based text extraction, data chunking, and indexing for retrieval.

5.1.4 Development Stage

The implementation phase focuses on developing the core system components and integrating them into a functional application. Backend services and APIs are built using Node.js and Express.js to handle secure document processing and AI-related operations. The database is implemented using local MongoDB, where data persistence and basic retrieval mechanisms are configured. The front-end interfaces are developed using React along with modern CSS frameworks to create dynamic User and Admin panels. Core features include integrating Tesseract.js for OCR processing, implementing a Gemini-based RAG pipeline, and developing structured legal advisory logic. Initial testing is conducted through unit testing of API endpoints and document processing functions to ensure system reliability.

5.1.5 Testing Stage

The testing phase focuses on validating system performance, accuracy, and reliability. Unit testing is conducted for individual modules to verify the accuracy of the Tesseract.js OCR engine and the relevance of the document retrieval mechanism. Integration and system testing ensure that administrative document updates are properly processed, stored, and retrieved by the user-facing AI system. User Acceptance Testing (UAT) is carried out using real-world land record samples and common legal queries in regional languages to evaluate practical usability. Identified bugs are fixed through iterative testing, with improvements made to server response time and prompt engineering for more accurate and reliable legal guidance.

5.1.6 Deployment Stage

The deployment phase focuses on preparing the system for execution and demonstration in a controlled environment. The application is configured to run on a local server using Node.js, ensuring all backend services and APIs function correctly. Third-party services such as Cloudinary and Google Gemini API are integrated using secure environment configurations. Initial datasets, including land laws, RERA

procedures, and Jantri-related information, are loaded into the system, and admin access is configured. Finally, the platform is prepared for demonstration and testing, allowing users to interact with the system in a local or controlled setup.

5.1.7 Maintenance Stage

The maintenance phase focuses on ensuring system reliability, performance, and continuous improvement. System performance and error logs are monitored to track API usage and evaluate the accuracy of Tesseract.js OCR processing. Support is provided to users and administrators by updating system data as land regulations and related documents evolve. Regular updates and bug fixes are applied to improve system stability and enhance AI response accuracy through prompt refinement and library updates. Periodic backups and basic security checks are performed to maintain data integrity and protect user-uploaded documents stored in MongoDB.

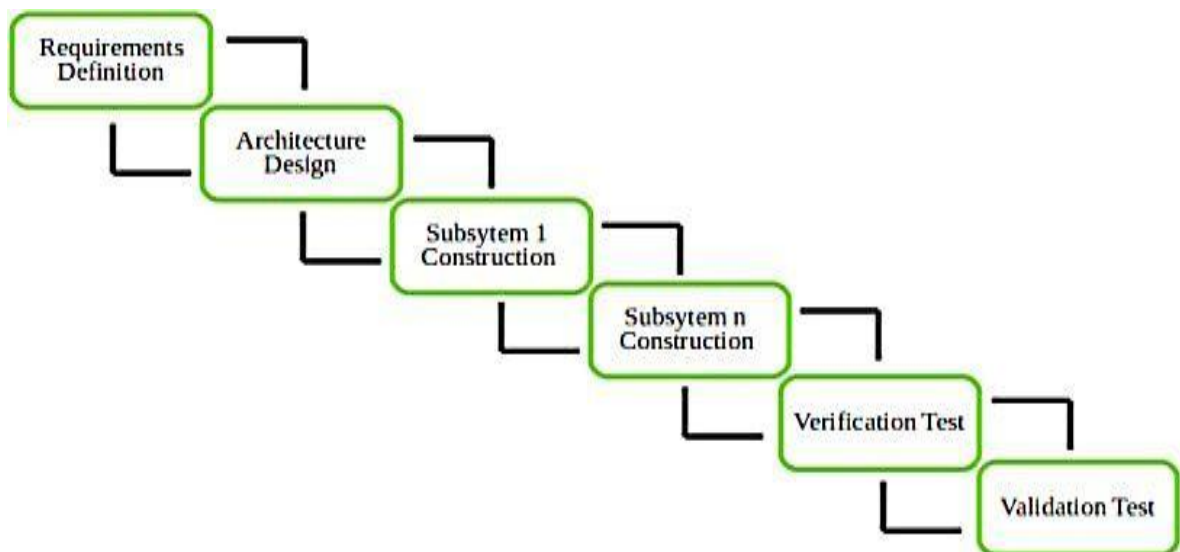


Figure 1 Incremental Model

Figure 1 represents a step-by-step software development process, starting from requirements definition and progressing through architecture design and subsystem construction. Each stage builds on the previous one, ensuring a structured and systematic development flow throughout the project lifecycle. Finally, the system undergoes verification and validation to ensure its correctness, reliability, and overall quality before deployment.

6. PROJECT PLANNING

This chapter outlines the project planning phase of the Ai Based Smart Land Document Analysis & Registry Assistant System focusing on defining objectives, scope, timelines, and resources. It highlights key considerations such as risk identification, technology selection, and development strategy. The planning ensures a clear roadmap for building a reliable and efficient system that meets user and technical requirements.

6.1 Project Planning

The planning phase of the AI Based smart land document analysis & registry assistant system project plays a crucial role in ensuring the smooth and successful development of the entire platform. During this phase, the project team prepares a comprehensive plan that defines the system's core objectives, such as AI-based interpretation of 7/12 Utara records, automated legal procedural advisory, and regional language OCR integration. The planning stage outlines system requirements, development timelines, optimized infrastructure costs, and potential risks involved in integrating Generative AI APIs, handling complex Gujarati script recognition, and managing sensitive land ownership data. A development duration of approximately 14 to 15 weeks is allocated, ensuring each module—including the RAG engine and the Admin verification portal—is implemented and tested systematically. This plan serves as a roadmap that guides every stage of the project and ensures that the team stays aligned with goals, timelines, and user expectations for legal accuracy.

Early identification of constraints such as API rate limits for Large Language Models, data consistency challenges in scanned revenue documents, and system scalability needs reduces the risk of delays or rework in later phases. This stage also includes risk assessment, resource planning, and technology selection (MERN stack, Gemini API, and Tesseract.js), laying a solid foundation for seamless execution. With well-defined planning, the project is positioned to deliver a reliable, user-friendly, and feature-rich platform that simplifies property-related decision-making.

6.2 Project Timeline

Activity	Description	Estimated Duration (Weeks)
Requirements Gathering	Gather and document functional and non-functional requirements, including indexing land records (7/12, Index-II), Gemini AI OCR capabilities, legal drafting rules, and data privacy protocols.	2 weeks
System Design	Design the system architecture, database schema (MongoDB), and UI/UX wireframes for the Unified Chat Interface, Document Analysis dashboard, and Voice Assistant overlay.	3 weeks
Frontend Development	Develop the web user interface using React.js and Vite, including document upload components, real-time analysis reports, voice interaction UI, and admin Audit Vault.	4 weeks
Backend Development	Implement server-side features including Google Gemini 2.5 Flash integration, Node.js/Express API, automated legal deed generation logic, and secure authentication.	5 weeks
Database Setup	Set up a scalable MongoDB database for storing user profiles, document metadata, verified/fraud flags, and persistent chat history.	2 weeks
Testing and Debugging	Test modules such as AI OCR accuracy, fraud detection algorithms, voice-to-speech consistency, and system performance under load. Ensure security checks for sensitive documents.	3 weeks
User Feedback and Refinement	Collect feedback on AI response accuracy and document analysis insights. Refine the legal drafting templates and optimize the chat interaction flow.	2 weeks
Final Testing and Deployment	Perform final end-to-end system testing and deploy the application to cloud servers (e.g., Vercel/Render) with secure API management and full functionality.	2 weeks
Documentation and Training	Prepare comprehensive project reports, user manuals, and Viva preparation guides for presenting the system to evaluators.	1 week
Maintenance and Updates	Provide ongoing support, optimize AI model prompts, fix edge-case bugs, and roll out feature enhancements based on evolving land record formats.	Ongoing

Table 3 Project Timeline

Table 1 presents the key project development activities along with their descriptions and estimated durations. It provides a structured overview of each phase, starting from requirements gathering to maintenance and updates. The table helps in understanding how the project is planned and executed step by step. It also ensures proper time management and coordination between different development stages. Overall, it reflects a systematic approach to completing the project efficiently.

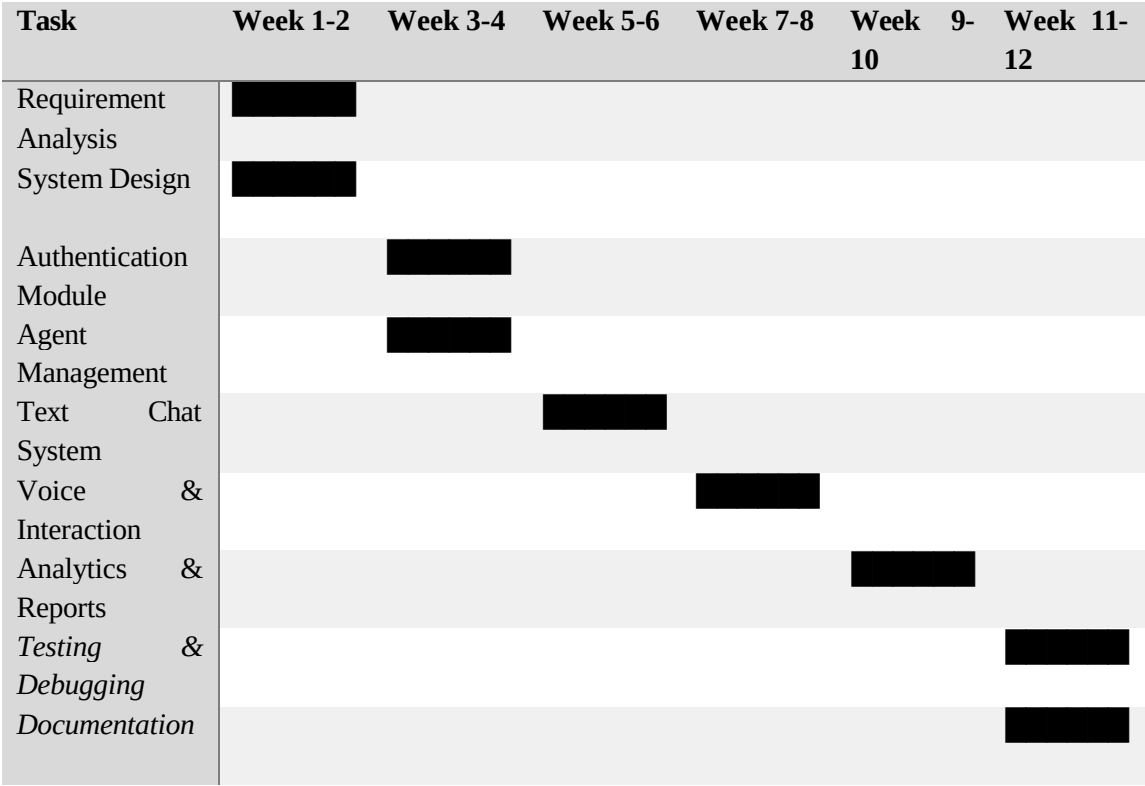


Figure 2 Gantt Chart

Figure 2 Project Timeline shows a Gantt chart representing the scheduling of various project activities over a defined time period. It highlights how tasks like requirements gathering, design, development, testing, and deployment are planned sequentially and sometimes overlap. This timeline helps in tracking progress and ensures systematic and timely completion of the project.

7. SYSTEM DESIGN

This chapter presents the UML approach used to model the AI Based Smart Land Document Analysis & Registry Assistant System. It explains how different diagrams are used to represent the system's structure, behavior, and data flow. These models help in clearly understanding system design, interactions, and database organization before actual implementation.

7.1 UML APPROACH

A UML (Unified Modeling Language) diagram is a graphical representation of a collection of elements within a system while UML diagrams can theoretically include any combination of elements and relationships, certain standard patterns tend to emerge in practice. These common combinations correspond to five key architectural views that are particularly useful for understanding complex, software-intensive systems. To support these views, the UML defines nine standard diagram types, each tailored to highlight specific aspects of the system, such as its static structure, dynamic behavior, or interactions among components. For this reason, the UML includes nine such diagrams:

- **Use-Case Diagram:** A use case is a description of how a user interacts with a system to achieve a specific goal or perform a task.
- **Sequence Diagram:** A sequence diagram is a UML diagram that shows the order of interactions between objects in a system over time.
- **Activity Diagram:** An activity diagram is a UML diagram that represents the flow of control or data through a system, showing the sequence of activities and decision points.
- **State Diagram:** A state diagram is a UML diagram that illustrates the various states an object can be in and the transitions between those states based on events or conditions.
- **Data Flow Diagram:** A Data Flow Diagram (DFD) is a visual representation that shows the flow of data within a system, illustrating how inputs are processed to produce outputs through various processes.
- **Class Diagram:** A class diagram is a UML diagram that shows the static structure of a system by depicting its classes, attributes, methods, and the relationships between classes.
- **ER Diagram:** A diagram that shows the system's main data entities and how they are related. It represents tables, their attributes, and the relationships between them in the database
- **Data Dictionary:** A document that defines all data fields used in the system, including their names, types, sizes, and descriptions. It provides a clear reference for understanding the structure of the database.

7.1.1 Use-Case Diagram

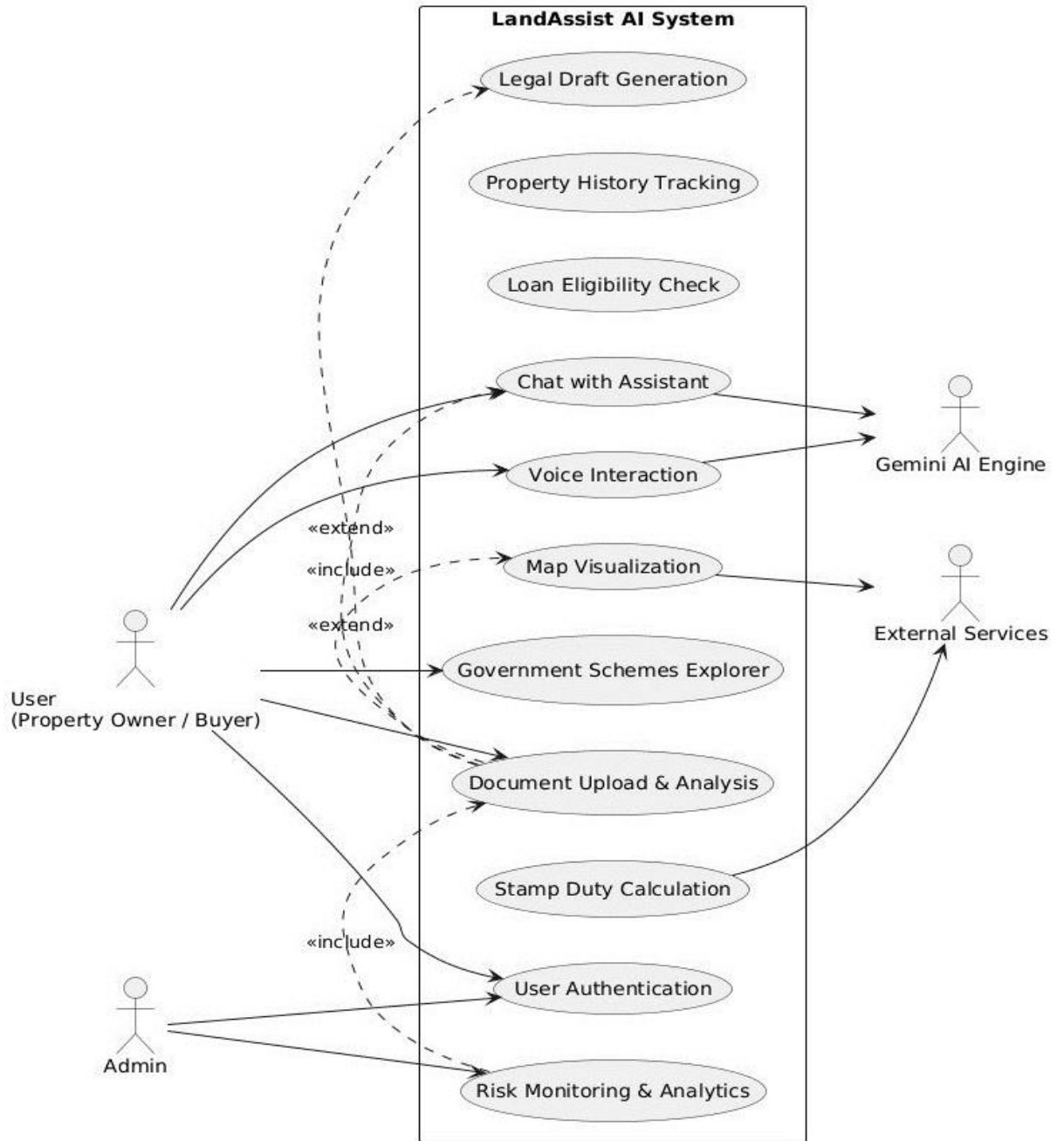


Figure 3 Use-Case Diagram

This diagram shows the interaction between users, admin, and the AI-Based Smart Land Document Analysis & Registry Assistant System, as shown in Figure 3. Users can upload documents, interact with the AI through chat, and access services such as loan eligibility checks and scheme exploration. The admin is responsible for authentication management and monitoring system activities. The system also integrates with external AI and service modules to process data and generate intelligent outputs

7.1.2 Sequence Diagram

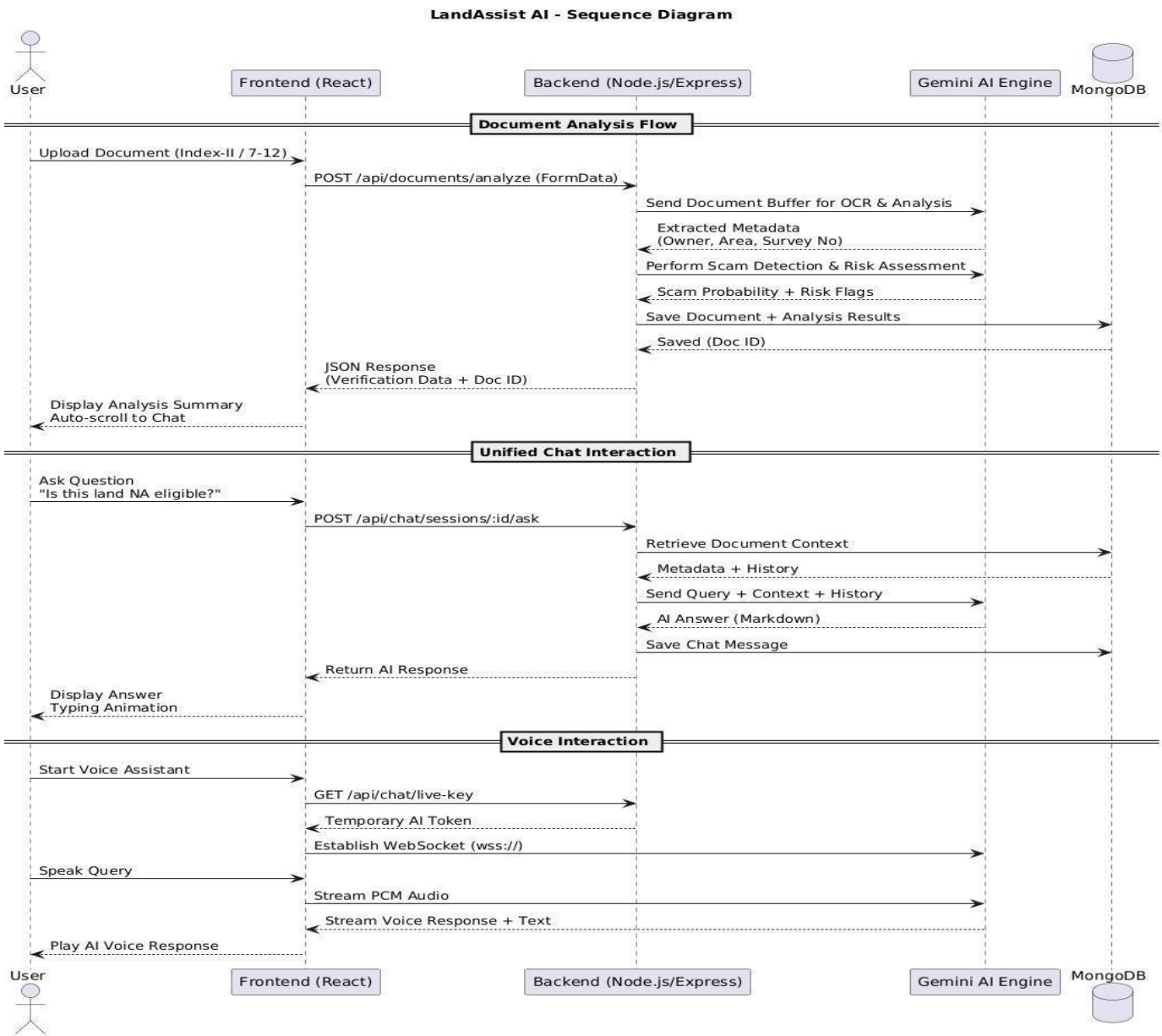


Figure 4 Sequence Diagram

This sequence diagram as shown in Figure 4 illustrates how the AI Based Smart Land Document Analysis & Registry Assistant System processes document analysis, chat interaction, and voice communication. The user uploads a document, which is processed through the backend and AI engine for OCR, risk analysis, and storage. In chat interaction, user queries are combined with document context to generate AI responses. The voice interaction flow enables real-time communication using audio streaming and AI-generated voice replies.

7.1.3 Activity Diagram

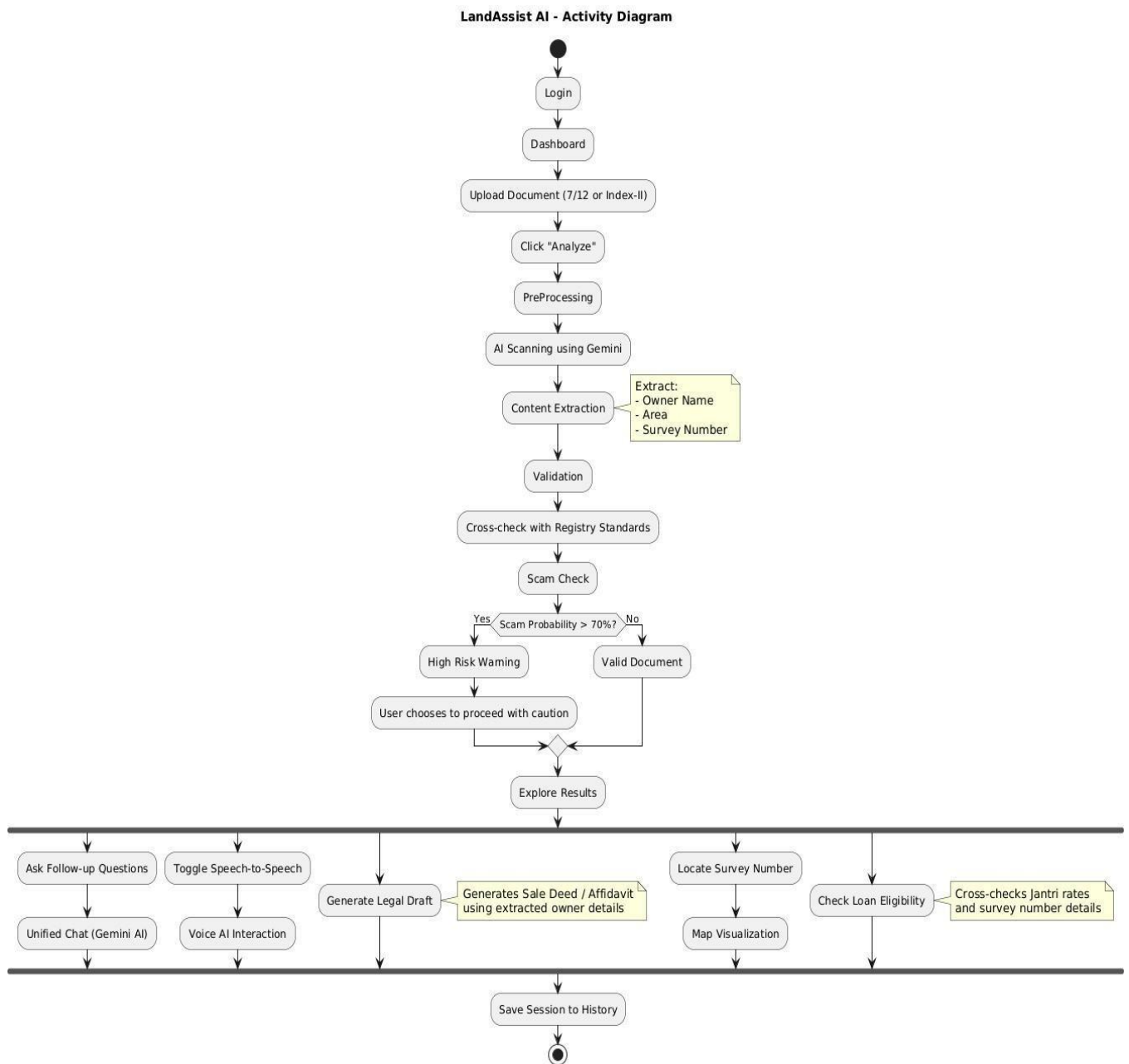


Figure 5 Activity Diagram

This activity diagram as shown in Figure 5 represents the workflow of the AI Based Smart Land Document Analysis & Registry Assistant System, starting from user login to document analysis and result generation. It shows how the system processes uploaded land documents through AI scanning, validation, and scam detection. Based on the analysis, users can explore results, generate legal drafts, or perform additional actions like chat, voice interaction, and map visualization.

7.1.4 Class Diagram

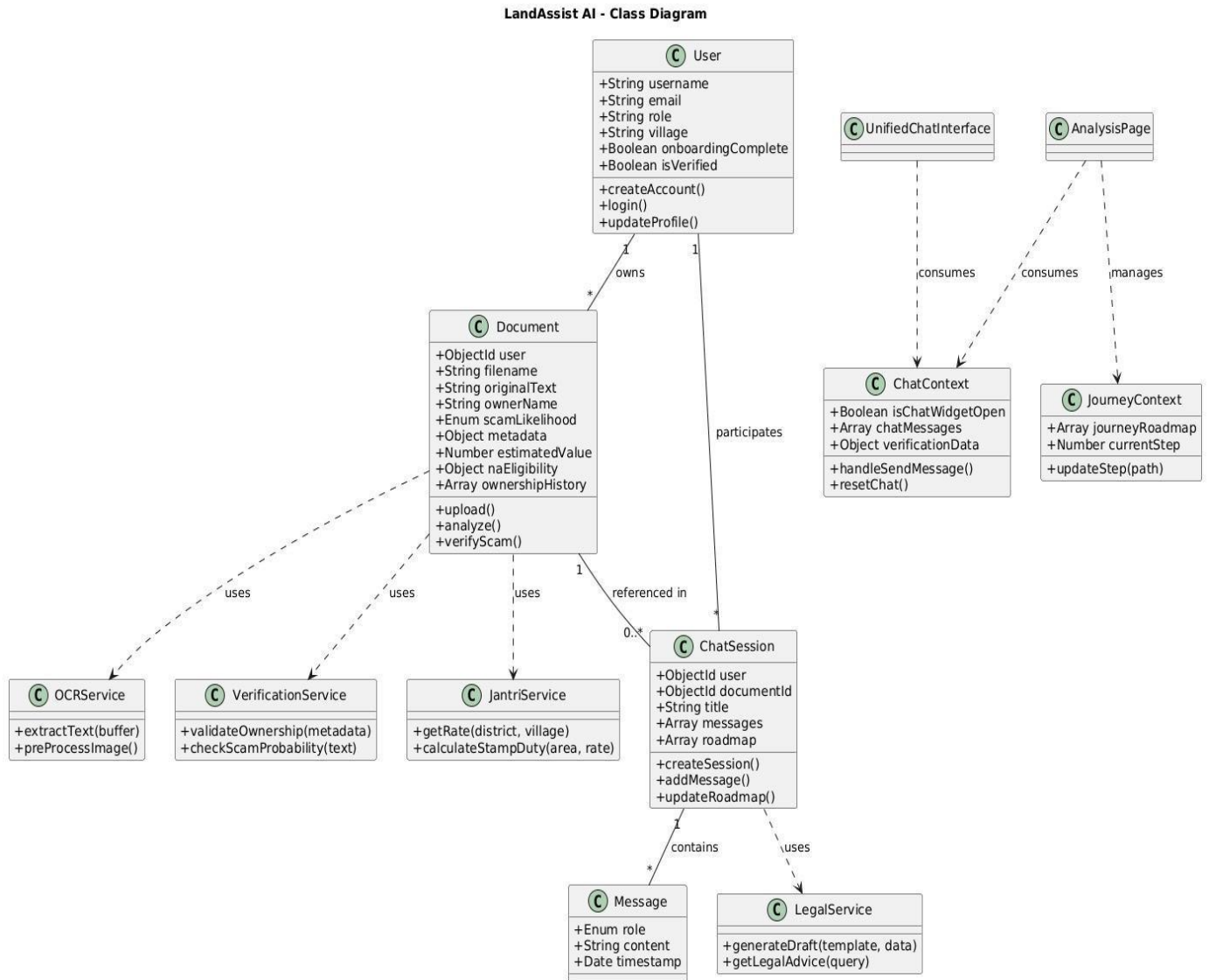


Figure 6 Class Diagram

This class diagram as shown in Figure 6 represents the structure of the AI Based Smart Land Document Analysis & Registry Assistant System by defining key classes and their attributes. It includes user-related details such as username, email, role, and verification status, along with methods like account creation and login.

7.1.5 State Diagram

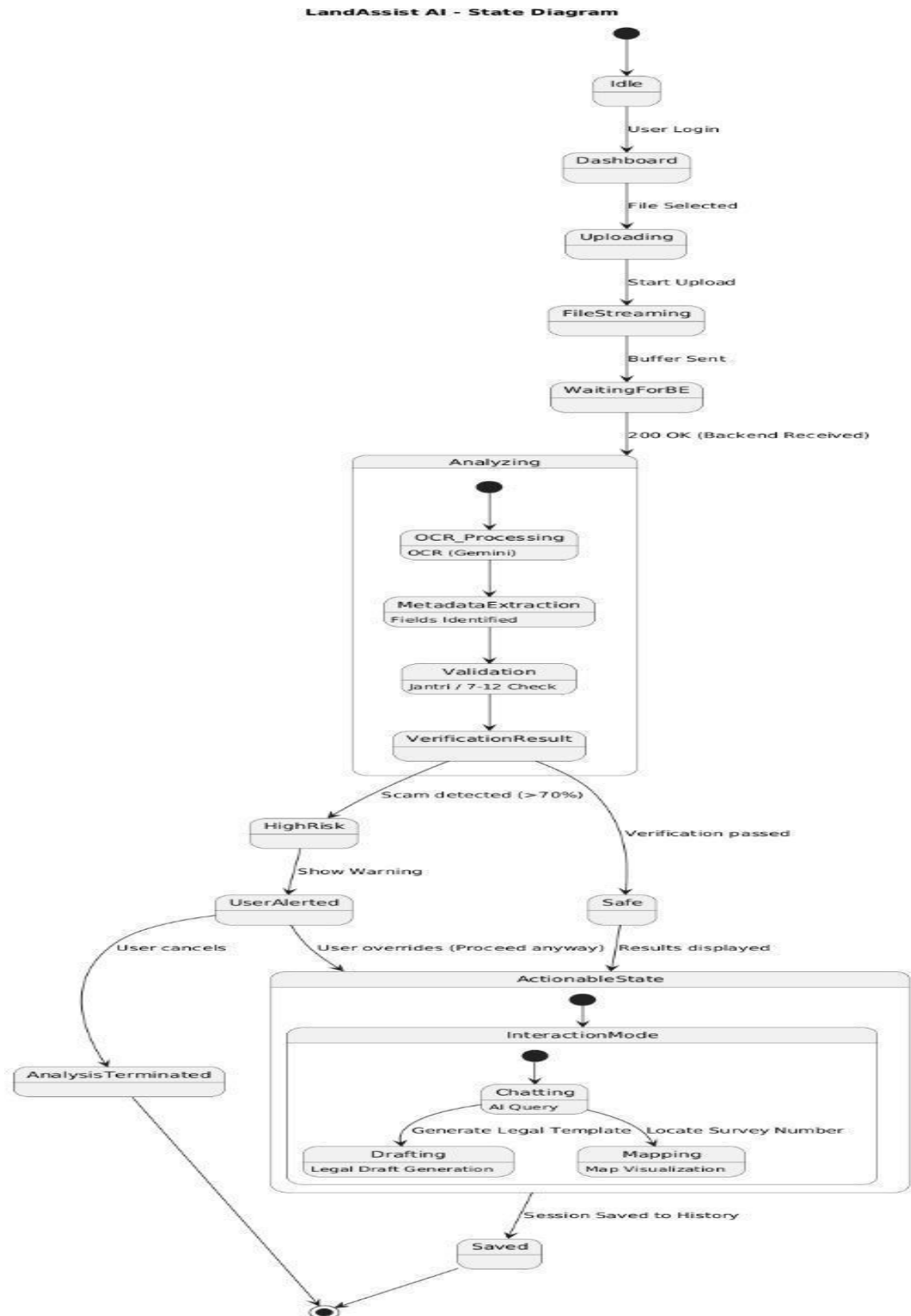


Figure 7 State Diagram

Figure 7 shows the state diagram of the LandAssist AI system, representing the complete workflow from user login to final output. It begins with file upload and moves through stages like file streaming, backend processing, and AI-based analysis including OCR, metadata extraction, and validation. Based on the verification result, the system identifies whether the document is safe or high risk and alerts the user accordingly. Finally, it enters an actionable state where users can interact, generate legal drafts, or view mapping results, and the session is saved for future reference.

7.1.6 Data Flow Diagram

The Data Flow Diagram (DFD) represents how data moves through the system, showing the interaction between users, processes, and data storage. It clearly illustrates how input data is processed and transformed into output, ensuring a proper understanding of system functionality. This helps in identifying data handling, processing steps, and overall system workflow.

7.1.6.1 Context Level/ Level 0

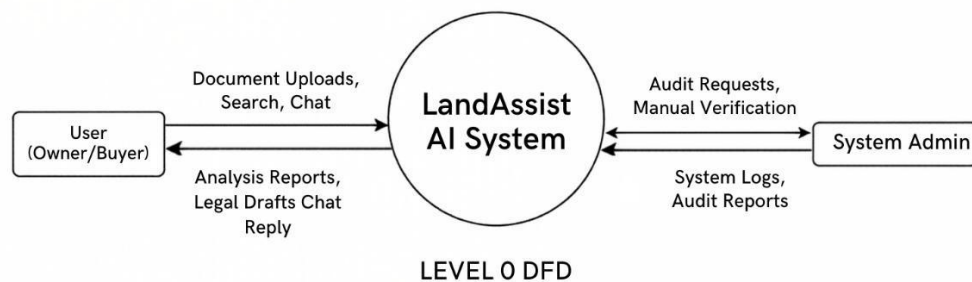
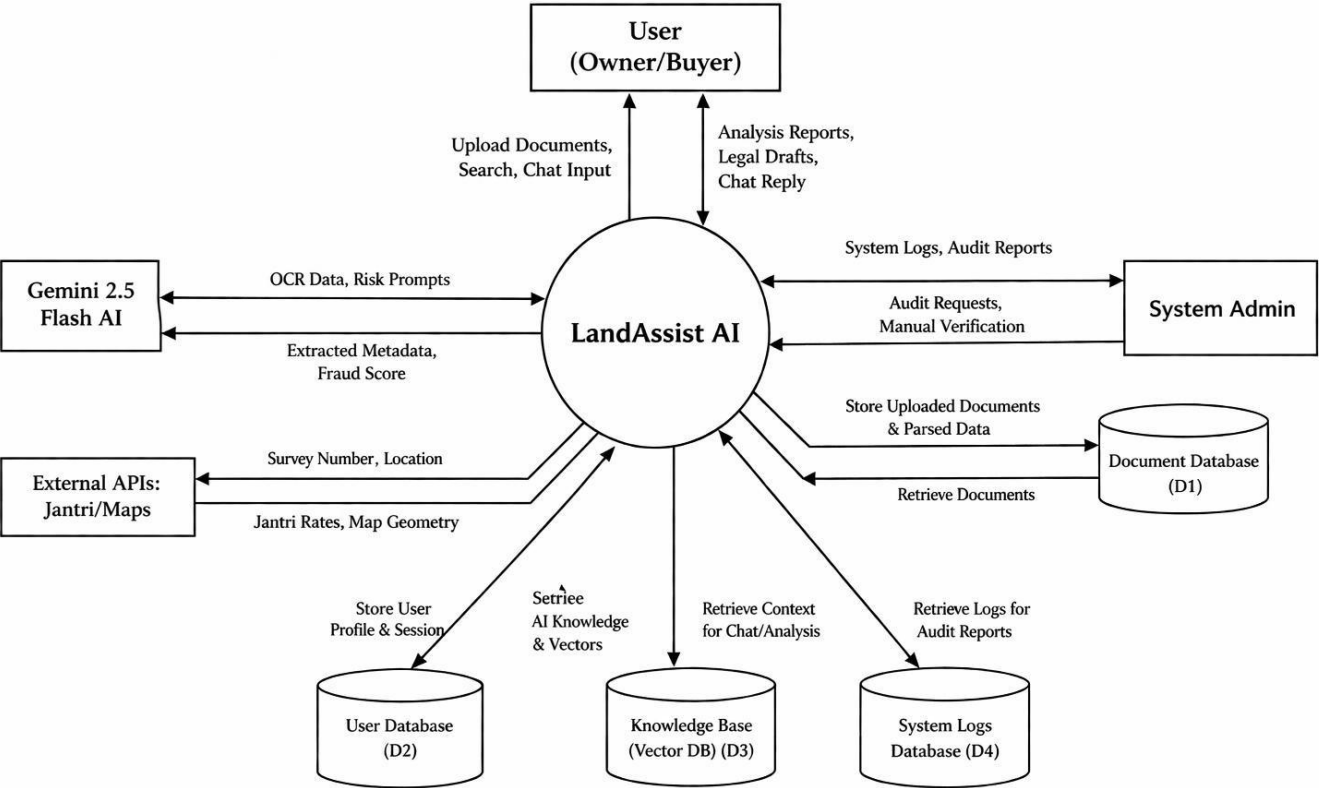


Figure 8 DFD Level 0

This Level 0 DFD represents the overall working of the Land Assist AI System, showing interactions between the User (Owner/Buyer) and the System Admin. The user provides inputs like document uploads, search queries, and chat requests, while the system processes them and returns analysis reports, legal drafts, and chat responses. The admin interacts with the system for audit requests and verification, and receives system logs and audit reports, ensuring proper monitoring and control of the system.

7.1.6.2 Level 1



DATA FLOW DIAGRAM: LANDASSIST AI (DFD LEVEL 1)

Figure 9 DFD Level 1

Figure 9 shows the DFD Level 1 of the AI-Based Smart Land Document Analysis & Registry Assistant System. It outlines the overall workflow where user-uploaded documents and admin-managed regulations are processed by the Gemini AI engine to generate risk analysis reports and interactive chat-based insights. The diagram highlights the flow of data from secure input to intelligent processing and output. It also represents how the system provides real-time assistance and accurate land-related information to the end user in a seamless manner.

7.1.6.3 Level 2

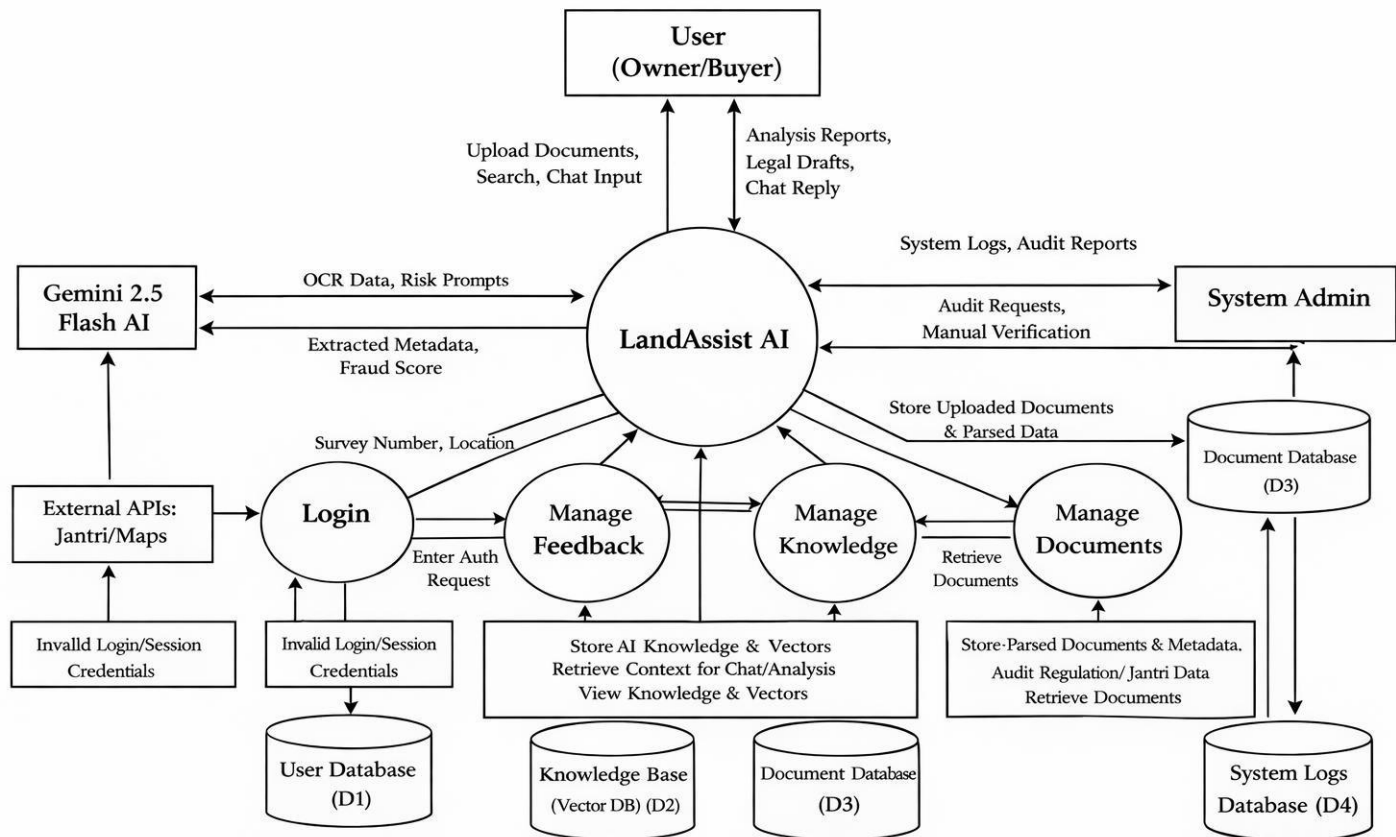


Figure 10 DFD Level-2

Figure 10 shows the DFD Level-2 of the AI-Based Smart Land Document Analysis & Registry Assistant System. It illustrates how the system processes land-related documents through multiple stages, including OCR-based text extraction, semantic analysis, and risk detection to ensure accuracy and reliability. The diagram provides a detailed view of internal processes and data flow within the system. It ultimately leads to the generation of a structured and comprehensive report, helping users make well-informed decisions efficiently.

7.1.7 ER Diagram

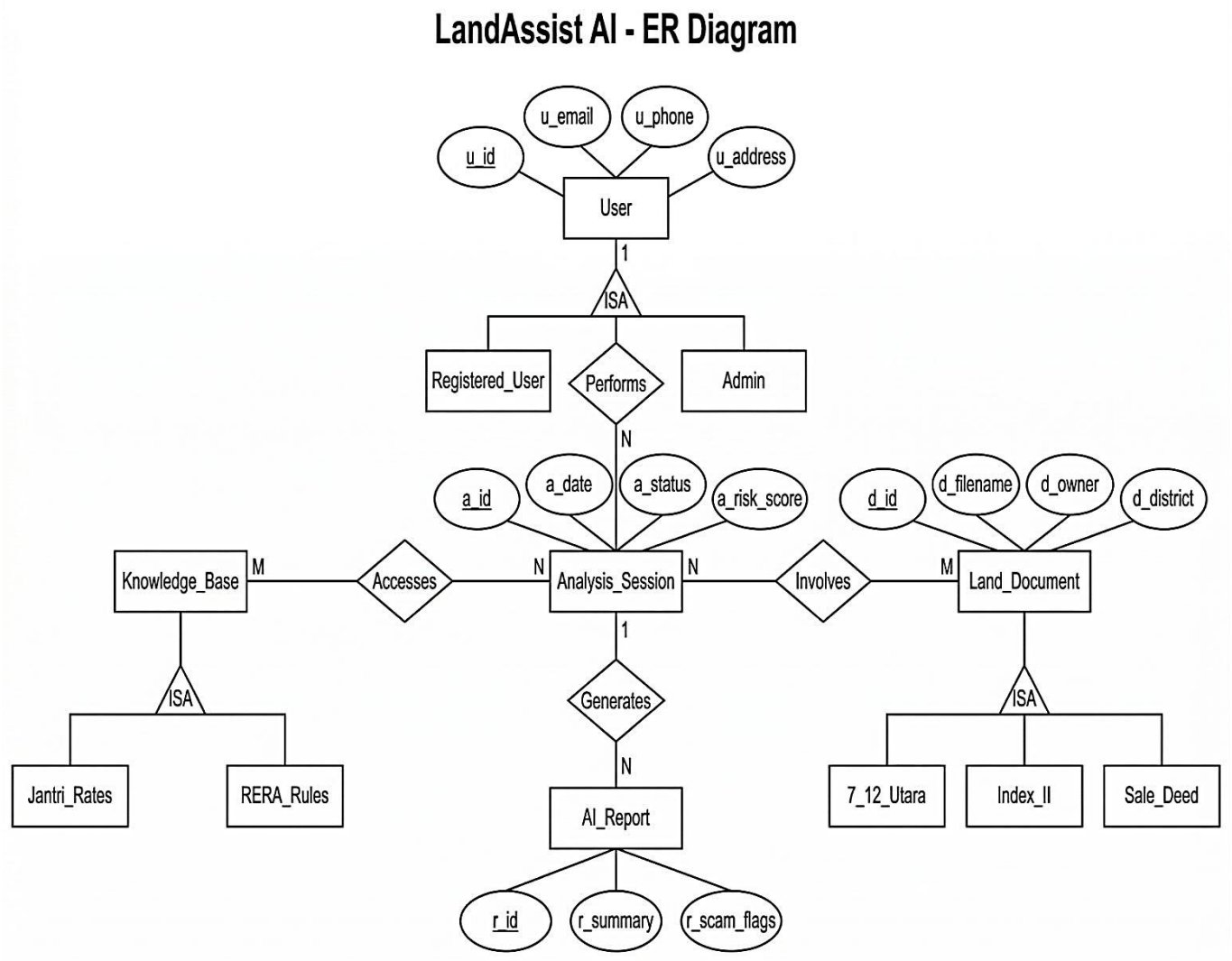


Figure 11 ER Diagram

Figure 11 shows the ER diagram of the system, illustrating how users initiate analysis sessions that interact with specific land documents such as Sale Deeds and Index II. These sessions are linked with a regulatory knowledge base to ensure accurate validation and processing of information. The diagram clearly represents the relationships between users, documents, and analysis components. It ultimately leads to the generation of AI reports that provide critical outputs such as risk scores, summaries, and scam detection flags, helping users make informed and secure decisions.

7.2 Data Dictionary

The Data Dictionary is a structured description of all the data elements used in the system. It defines the meaning, type, format, and purpose of each data field to ensure consistency and clarity across the platform. It acts as a central reference for understanding how data is organized, stored, and maintained within the system. It includes details such as data types (e.g., integer, string, date), field sizes, default values, and constraints, which help maintain data integrity and standardization. By clearly defining each data element, it reduces ambiguity and ensures that all developers and stakeholders interpret the data in the same way.

7.2.1 Users Table

Field Name	Data Type	Description	Constraints
_id	Objected	Unique identifier for the user.	Primary Key
username	String	Unique handle for the user.	Unique, Sparse
email	String	User's email address for login.	Required, Unique
password	String	Hashed password.	Required (if not Google)
googled	String	ID for Google OAuth users.	Sparse
role	String	User role (user/admin).	Default: 'user'
phone	String	User's contact number.	Optional
district	String	User's base district.	Optional
is Verified	Boolean	Account verification status.	Default: false
created at	Date	Account creation timestamp.	Default: Date Now

Table 4 Users Table

This table outlines the database structure for user accounts, defining how login credentials, profile details, and account status are stored, as shown in Table 4. It ensures

each user has a unique identity and an assigned role (user or admin) to maintain secure access to the platform.

7.2.2 Document Table

Field Name	Data Type	Description	Constraints
_id	Objected	Unique identifier for the document.	Primary Key
user	Objected	Reference to the User who uploaded.	Foreign Key, Required
filename	String	Original name of the file.	Required
original Text	String	Extracted text from OCR.	Required
owner Name	String	Name of the owner parsed from AI.	Optional
scam Likelihood	String	AI-calculated risk (Low/Medium/High).	Default: 'Low'
jantri Rate	Number	Calculated market rate per unit.	Optional
metadata	Object	Parsed data (Survey No, Area, etc.).	Optional
uploaded at	Date	Document upload timestamp.	Default: Date. Now

Table 5 Document Table

The Document Table defines the schema for managing uploaded files, capturing essential metadata such as user ownership, filenames, and timestamps, as shown in Table 5. It is designed to store both raw OCR-extracted text and AI-generated insights, including scam likelihood assessments and market rate calculations, to ensure structured and efficient data retrieval. Additionally, the table maintains document status indicators such as verification results, processing stages, and risk flags to support accurate tracking and monitoring. It also enables linkage with related entities like user profiles and analysis sessions, ensuring seamless data integration across the system. Proper indexing and organization of fields improve query performance, making it faster to access and analyze documents. Overall, the table plays a critical role in maintaining

data consistency, supporting advanced analysis, and ensuring reliable document management within the platform.

7.2.3 Chat Session Table

Field Name	Data Type	Description	Constraints
_id	Objected	Unique identifier for the session.	Primary Key
user	Objected	Reference to the User.	Foreign Key, Required
documented	Objected	Reference to the document in context.	Foreign Key, Optional
title	String	Auto-generated title for the chat.	Default: 'New Chat'
messages	Array	List of message objects (role, content).	Required
updated At	Date	Last interaction timestamp.	Default: Date. Now

Table 6 Chat Session Table

The Chat Session Table is designed to manage individual conversation logs by linking users to specific documents for context-aware interactions, as shown in Table 6. It stores an array of message objects alongside auto-generated titles and timestamps, ensuring that user chat histories are organized and easily retrievable.

7.2.4 Tracks legal tasks and property

Field Name	Data Type	Description	Constraints
_id	Objected	Unique identifier for the task.	Primary Key
user	Objected	Reference to the User.	Foreign Key, Required
title	String	Short description of the task.	Required
due Date	Date	Deadline for the task.	Required

Table 7 Tracks legal tasks and property

The Tracks Legal Tasks and Property Table serves as a dedicated management schema for organizing user-specific legal obligations and deadlines, as shown in Table 7. It captures essential metadata, including task priority and completion status, to ensure efficient monitoring and timely execution of property-related responsibilities.

7.2.5 Audit Log Table

Field Name	Data Type	Description	Constraints
_id	Objected	Unique identifier for the log entry.	Primary Key
type	String	Type of log (login, document, admin).	Required
user	Objected	Reference to the performing user.	Foreign Key, Optional
action	String	Description of the action performed.	Required
Ip Address	String	IP of the request.	Default: 'unknown'
created at	Date	Time of the action.	Default: Date. Now

Table 8 Audit Log Table

The Audit Log Table serves as a centralized security and activity tracker, recording every significant action performed by users or admins within the system, as shown in Table 8. It captures vital details such as the action type, originating IP address, and exact timestamps to ensure accountability and facilitate system monitoring or debugging.

7.2.6 Rule Document Table

Field Name	Data Type	Description	Constraints
_id	Objected	Unique identifier for the rule.	Primary Key
title	String	Title of the legal document/rule.	Required
file Path	String	Path to the stored PDF/Doc file.	Required
district	String	District the rule applies to.	Optional
uploaded at	Date	Creation timestamp.	Default: Date. Now

Table 9 Rule Document Table

The Rule Document Table acts as a repository for official legal guidelines and regulatory documents, storing their file paths and descriptive titles, as shown in Table 9. It supports region-specific filtering by associating rules with particular districts, ensuring that users have access to the correct legal context for their location. In addition, the table maintains versioning and update information to keep track of changes in regulations over time. This ensures that the system always uses the most recent and relevant legal rules during document analysis. It also allows administrators to easily add, modify, or remove rules, making the system flexible and adaptable to policy changes. Overall, the table plays a key role in enhancing the accuracy and reliability of AI-driven evaluations by providing a well-structured and up-to-date legal knowledge base.

7.3 UI/UX Screens

UI/UX Screens presents the visual interface design of the system, showing how users interact with different modules of the application. It includes key screens such as dashboards, document upload interfaces, analysis reports, and administrative panels. These screens are designed with a clean, intuitive, and user-friendly layout to ensure smooth navigation and ease of use. The section highlights how the UI/UX design improves overall user experience and makes the system more efficient and accessible.

7.3.1 Login

The Login module is a fundamental part of the system that handles user authentication and secure access control. It requires users to enter valid credentials such as username/email and password. These credentials are verified against the stored records in the database. If the credentials match, the system grants access and redirects the user to the dashboard based on their role. If the credentials are incorrect, the system denies access and displays an appropriate error message. This module helps in preventing

unauthorized access to sensitive system data. It also ensures that only registered users can interact with the application. attempts to enhance security. The Login module also maintains user session integrity during system usage. It acts as the first layer of defense for the entire application. Overall, it plays a critical role in ensuring system security, data protection, and controlled user access.

The screenshot shows the login interface of the GovTech AI Gujarat Land Records Portal. At the top, the header includes the portal's name and navigation links: Home, About AnyRoR, Help, and a Help Desk button. The main content area is a light blue box containing a white login card. The card starts with a fingerprint icon and a 'Welcome Back' greeting, followed by the instruction 'Access your official land document analysis dashboard'. The login fields include 'Email or Phone Number' with a placeholder 'Enter registered email/phone' and a 'Password' field with a placeholder '*****'. A 'Sign In' button with a right-pointing arrow is located below the password field. A link for 'I forgot Password?' is positioned to the right of the password field. Below the login card, there is a section for 'OR LOGIN w/ ITH' with two buttons: 'AI P' and 'Google'. A security notice at the bottom of the card reads: 'Authorized Access Only. Secured by the Revenue Department, Government of Gujarat. your session is encrypted and monitored for security.' At the very bottom of the card, there is a link: 'Don't have an account? Create Account'. The footer of the page contains three status indicators: 'SSL ENCRYPTED', 'GOVERNMENT VERIFIED', and '24/7 MONITORING'.

Figure 12 Login Page

7.3.2 Dashboard Screen

The Dashboard screen is the main interface of the system after successful login. It provides a centralized view of all key features and modules available to the user. The dashboard displays quick access options such as document upload, analysis reports, history, and profile settings. It also shows important summaries like recent activities, status updates, or notifications. The layout is designed to be simple, user-friendly, and easy to navigate. Based on user roles, different dashboard elements may be shown to ensure proper access control. The dashboard acts as a control center from which users can move to other parts of the system efficiently. It improves usability by reducing navigation time and organizing features in a structured manner.

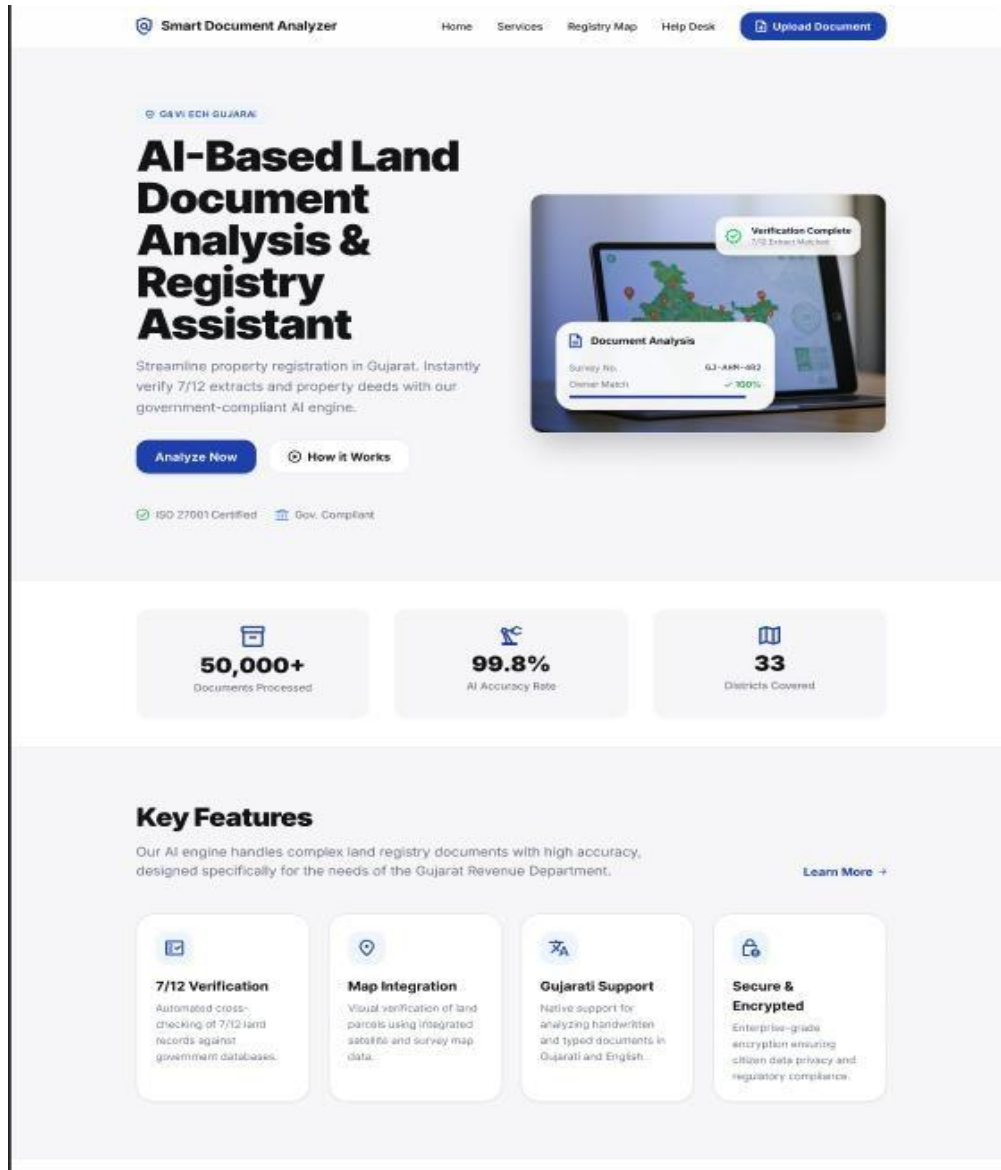


Figure 13 Dashboard Screen

7.3.3 Jantri & Stamp Duty Calculator

The Jantri & Stamp Duty Calculator module is used to estimate property value and applicable stamp duty charges based on government-defined rates. Users can input property details such as location, area, and type of land or property. The system then uses predefined Jantri rates to calculate the approximate market value of the property. Based on this value, the stamp duty is computed according to applicable legal percentages. This module helps users understand the financial implications of property transactions. It reduces manual calculation errors and improves accuracy in valuation. The interface is designed to be simple so that users can easily enter data and get instant results. It is especially useful for buyers, sellers, and legal verification processes in land-related transactions.

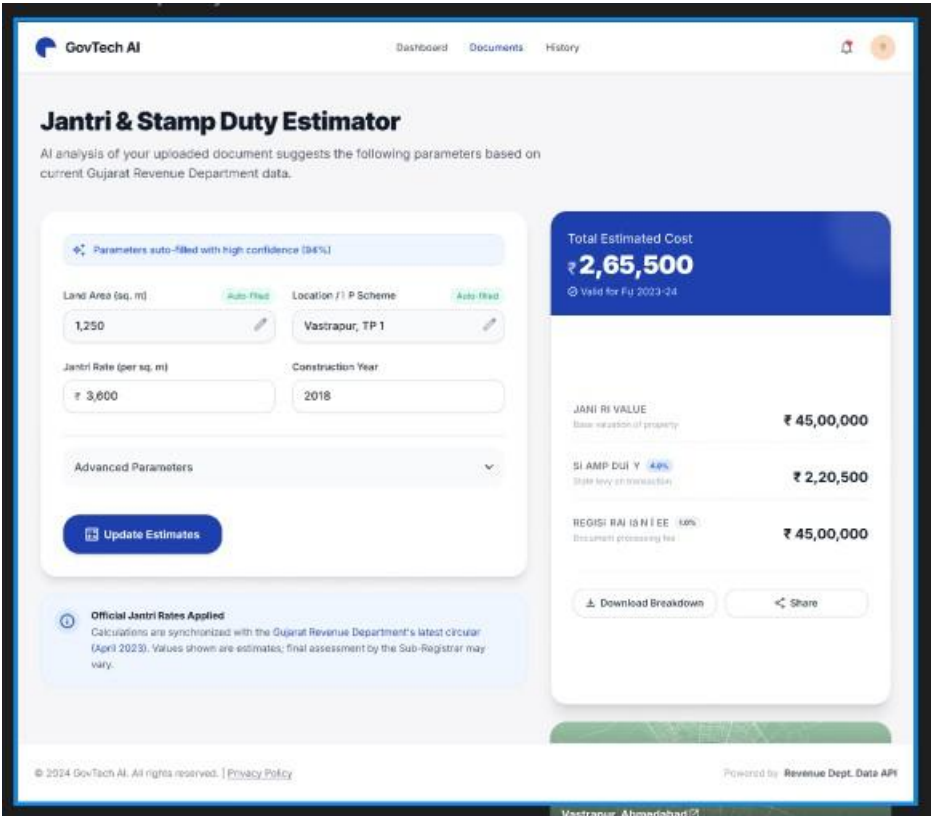


Figure 14 Jantri & Stamp Duty Calculator

8. IMPLEMENTATION DETAILS

This chapter provides a detailed explanation of how the Land Document Analysis & Registry Assistant System is implemented in practice. It describes the system architecture, including both backend and frontend components, and explains how different modules interact with each other. The chapter also highlights the key algorithms and techniques used in the system, showing how the design is converted into a working system step by step.

8.1 Backend Architecture

The backend is implemented in Node.js 18+ using the Express.js 5.0 web framework. The application follows a modular service-oriented architecture, organized into several key modules: `server.js` (main Express application and middleware), `geminiFileService.js` (Google Gemini File API integration), `jantriService.js` (RAG-based valuation extraction), `reportService.js` (PDF report and legal draft generation), and `geocodeService.js` (OSM Nominatim integration for land mapping). The Valuation Engine (`jantriService.js`) is the core intelligence layer for financial assessment. It utilizes a Retrieval-Augmented Generation (RAG) approach where official government Jantri PDF documents are indexed within the Gemini vector space. When a user uploads a document, the engine identifies the district and village, retrieves the corresponding Jantri PDF, and performs a targeted semantic search to extract the exact rate per square meter, ensuring high accuracy without a manual database of millions of rates.

8.2 Frontend Architecture

The frontend is a Single Page Application (SPA) built with React 19, Vite, and Tailwinds 4. It leverages a component-based architecture for high reusability. Navigation is managed by React Router, while real-time communication for document processing updates is handled via Socket.io-client. The UI is enhanced with Framer Motion for smooth transitions and Lucide-react for professional iconography. The Interactive Land Visualizer is implemented using Leaflet.js. When document metadata is extracted, the `geocodeService.js` resolves the village and taluka coordinates. The frontend then renders a dynamic map with spatial markers, allowing users to verify the physical location of the land parcel. Financial data is visualized using Recharts, providing users with a graphical breakdown of land value trends and comparative.

8.3 Key Algorithms

BEGIN

1. // ----- System Initialization -----
2. LOAD AI_Model (Gemini API)
3. LOAD OCR Engine
4. CONNECT Document_Database
5. CONNECT Knowledge_Base (Vector DB)
6. // ----- Document Upload & Preprocessing -----
7. FUNCTION process_document(user_document)
8. IF document_format != text THEN
9. extracted_text ← OCR_Engine.extract(user_document)

```
10. ELSE
11.     extracted_text ← user_document
12. ENDIF
13. cleaned_text ← preprocess(extracted_text)
14. RETURN cleaned_text
15. END FUNCTION
16. document_text ← process_document(user_document)
17. // ----- Metadata Extraction -----
18. metadata ← extract_entities(document_text)
19. survey_no ← metadata.survey_number
20. owner_details ← metadata.owner_info
21. location ← metadata.location
22. // ----- External Validation -----
23. jantri_data ← call_API(Jantri/Maps, survey_no, location)
24. validation_result ← compare(metadata, jantri_data)
25. // ----- AI Risk Analysis -----
26. risk_prompt ← build_prompt(document_text, metadata, validation_result)
27. ai_analysis ← call_Gemini_API(risk_prompt)
28. fraud_score ← extract_fraud_score(ai_analysis)
29. risk_flags ← extract_risk_points(ai_analysis)
30. // ----- Store Data -----
31. STORE document_text INTO Document_Database
32. STORE metadata, fraud_score INTO Database
33. // ----- Report Generation -----
34. analysis_report ← generate_report(metadata,
                                   validation_result,
                                   fraud_score,
                                   risk_flags)
35. // ----- Chat Assistant -----
36. FUNCTION chat_assistant(user_query)
37.     context ← retrieve_from_Knowledge_Base(user_query)
38.     chat_prompt ← combine(context, user_query)
39.     response ← call_Gemini_API(chat_prompt)
40.     RETURN response
41. END FUNCTION
42. IF user_query EXISTS THEN
43.     AI_response ← chat_assistant(user_query)
44. ENDIF
45. // ----- Output -----
46. RETURN analysis_report, fraud_score, AI_response
END
```

8.4 FlowChart

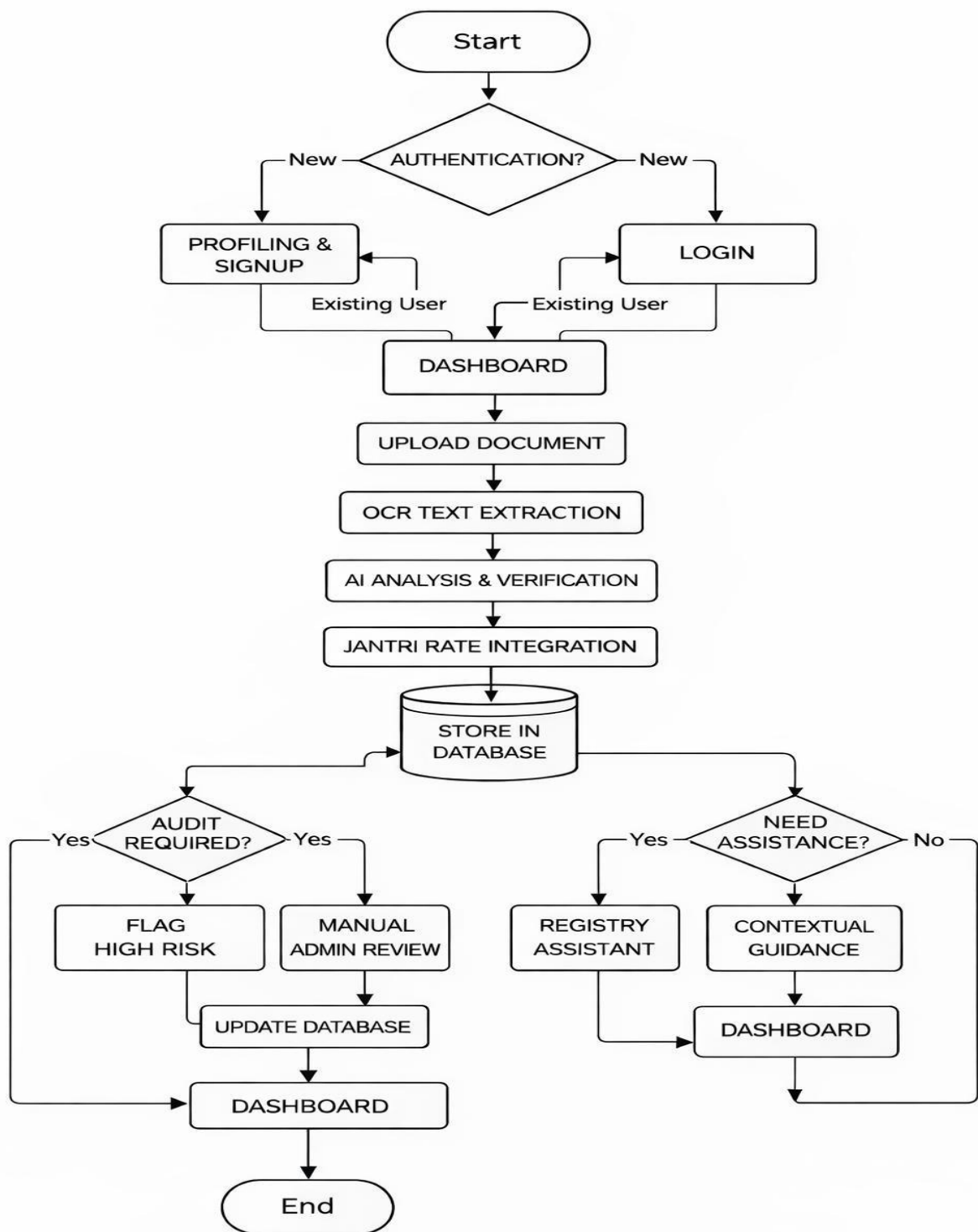


Figure 15 FlowChart

The figure shows the workflow diagram of the AI-Based Smart Land Document Analysis & Registry Assistant System, starting with user authentication where users sign up or log in to access the platform. After authentication, users are directed to the dashboard where they upload land-related documents for processing.

QR Code:

Figure 16 QR Code of GitHub

Figure 16 shows the QR code representation of the AI-Based Smart Land Document Analysis & Registry Assistant System link. This QR code allows users to quickly access the application or demonstration page using a mobile device, improving accessibility and ease of sharing.

9. TESTING

The testing of the AI-Based Smart Land Document Analysis Registry Assistant System was carried out using an iterative and incremental approach aligned with Agile methodology. Testing was performed continuously during development to identify and fix issues at an early stage. Various testing levels, including unit, integration, system, and user acceptance testing, were conducted to ensure the system's reliability and performance. This approach helped in delivering a stable and user-friendly application.

9.1 Testing Model Used

For the AI-Based Smart Land Document Analysis & Registry Assistant System, an Iterative and Incremental Testing Approach aligned with Agile methodology was utilized. Instead of restricting testing to the final phase, it was performed continuously throughout development to ensure early detection of errors and faster resolution. This helped in improving code quality, reducing integration issues, and maintaining system stability across all modules. The testing process ensured that both functional and non-functional requirements were properly validated under different scenarios. It also helped in verifying data accuracy, system reliability, and performance consistency during document processing and AI-based analysis.

The comprehensive testing process included the following levels:

1. **Unit Testing:** Individual components such as the Document Parser (OCR), Stamp Duty Calculator (Estimator), Jantri Rate Evaluation module, and Authentication system were tested independently. Each module was validated for correctness, input handling, and error management to ensure reliable functionality at the component level.
2. **Integration Testing:** After unit testing, modules were combined and tested to verify proper interaction and data flow between them. This included testing communication between the React frontend and Node.js/Express backend, as well as ensuring accurate transmission of OCR-extracted data to the AI analysis engine without loss or distortion.
3. **System Testing:** The complete integrated application was tested as a single system to evaluate overall performance and behavior. This phase confirmed that all features worked together as expected, including document upload, AI processing, report generation, and database storage, under both normal and stress conditions.
4. **User Acceptance Testing (UAT):** End-user scenarios were simulated to validate usability and practicality of the system. Users tested key functionalities such as uploading land documents, viewing AI-generated analysis reports, checking valuation results, and downloading final reports in PDF format. Feedback was collected and used to refine UI/UX and improve system responsiveness.

9.2 Test Cases and Test Results

This section presents the test cases designed to validate the functionality and performance of the AI-Based Smart Land Document Analysis & Registry Assistant System. Each test case includes input conditions, expected outputs, and actual results to ensure the system operates as intended. The results confirm that key features such as document upload, OCR processing, AI-based analysis, risk detection, and report generation are functioning correctly. Any discrepancies identified during testing were analyzed and resolved to improve system reliability and accuracy.

9.2.1 Authentication & Access Control

TC ID	Module	Test Description/Scenario	Desired (Expected) Result	Actual Result	Status	How it was Solved (Resolution)
TC_AUTH_01	Auth	User Registration (Valid data)	Account created; email verification sent.	Account saved; email received	PASS	Implemented using Crypt hashing for password & Node mailer for email.
TC_AUTH_02	Auth	Login with incorrect password	System shows "Invalid credentials".	Error message displayed correctly.	PASS	Solved using Express error-handling middleware.
TC_AUTH_03	Auth	Password Reset (Invalid Email)	System shows "Email not found".	Correct error message shown.	PASS	Implemented lookup check in MongoDB User model.
TC_AUTH_04	Auth	Role-based Access (User to Admin)	User should not see "Audit Vault".	Navigation restricted to User features.	PASS	Solved using React Context user Role.

TC_AUTH_05	Auth	Session Timeout (Expired Token)	User should be logged out after 24h.	Token expired; redirect to Login.	PASS	Configured JWT expiresIn: '24h' in backend.
TC_AUTH_06	Auth	Empty Field Registration	Show "All fields are required" error.	Fields highlighted red.	PASS	Added Yup/Zod schema validation on Frontend and Join Backend.
TC_AUTH_07	Auth	Duplicate Email Registration	Show "Email already exists" error.	Error handled without crashing.	PASS	Handled via MongoDB unique: true index and try-catch block.

Table 10 Authentication & Access Control

The table shows testing of the Authentication module, covering features like user registration, login, password management, and security checks, as shown in Table 10. It compares expected and actual results for each test case. Most tests passed successfully, ensuring system reliability. Identified issues were resolved using proper techniques like encoding fixes and token management.

9.2.1 Document Upload & Storage

TC ID	Module	Test Description/Scenario	Desired (Expected) Result	Actual Result	Status	How it was Solved (Resolution)
TC_UPL_01	Upload	Upload valid Land Deed (PDF)	File uploaded; processing starts.	Upload successful; preview visible.	PASS	Solved using Multer (storage) and Cloudinary (hosting).
TC_UPL_02	Upload	Upload very large file (>100MB)	Show "File too large" error.	Server crashed/timed out.	FAIL	Solved by increasing Node.js client_max_body_size and adding Multer limits.
TC_UPL_03	Upload	Invalid File Type (SVG/EXE)	Reject upload with error message.	Error message: "Only PDF/JPG/PNG".	PASS	Added file filter in Multer configuration.
TC_UPL_04	Upload	Simultaneous Multiple Uploads	Handle multiple files in queues.	All files processed sequentially.	PASS	Implemented Promise.all for parallel Cloudinary uploads.
TC_UPL_05	Upload	Upload with Slow Internet	Handle timeouts gracefully.	Show progress bar / Retry option.	PASS	Added Axios progress event listener on Frontend.
TC_UPL_06	Upload	File Rename on Server	Sanitize file names (remove spaces).	File saved as user_id_document.pdf.	PASS	Customized Multer destination and filename functions.
TC_UPL_07	Upload	Delete Uploaded Document	Permanent removal from Cloudinary.	File removed and DB entry deleted.	PASS	Used cloudinary.uploader.destroy and Mongoose findOneAndDelete.

9.2.2 Document Upload & Storage

TC_UPL_08	Upload	Checksum Verification	Ensure file integrity after upload.	Hashes match after transfer.	PASS	Added SHA-256 hashing pre- and post-upload for critical docs.
TC_UPL_09	Upload	Metadata Extraction (Client side)	Extract file size/name before upload.	Metadata displayed in preview.	PASS	Used browser File API to show local details.
TC_UPL_10	Upload	Server Storage Full	Graceful error handling for disk space.	"Upload failed: Server error".	PASS	Added disk space check middleware in backend.
TC_UPL_11	Upload	Corrupted PDF File	System should reject corrupt binary.	Processing failed but gave no error.	FAIL	Solved by adding pdf-parse integrity check post-upload.
TC_UPL_12	Upload	Filename too long (>255 chars)	System should truncate/reject.	File system error (ENONENT).	FAIL	Solved by adding path length validation in Multer logic.

Table 11 Document Upload & Storage

The table represents testing of the Upload module, covering scenarios like file upload, validation, size limits, and error handling, as shown in Table 11. It compares expected and actual results to ensure proper system functionality. Most test cases passed successfully, confirming reliable upload performance. Some failures were identified and resolved using techniques like file validation, hashing, and improved backend handling.

9.2.3 AI-Powered OCR & Data Extraction

TC_OCR_01	AI OCR	Extraction of Owner Name	Extract full names (First/Last).	Name mapped to UserProfile.	PASS	Refined prompt to return JSON with specific key mapping.
TC_OCR_02	AI OCR	Date of Registration Extraction	Extract DD/MM/YYYY format.	Standardized date format.	PASS	Used Regex parsing on Gemini output to normalize dates.
TC_OCR_03	AI OCR	Survey Number Detection	Identify survey/plot numbers.	Extracted accurately.	PASS	Integrated custom "Search Pattern" for regional document styles.
TC_OCR_04	AI OCR	Multilingual Support (Hindi/Gujarati)	Extract text in non-English scripts.	Accurate translation/extraction.	PASS	Leveraged Gemini's native multilingual OCR capabilities.
TC_OCR_05	AI OCR	Multi-page PDF Processing	Analyze all pages of the document.	Data aggregated from 10 pages.	PASS	Implemented PDF splitting and sequential API calls for large docs.
TC_OCR_06	AI OCR	Error Handling (Garbled Text)	Identify if doc is fake or unreadable.	Flag "Unreadable document".	PASS	Added heuristic check for high confidence scores from AI.
TC_OCR_07	AI OCR	Extraction Latency	Response should be within 10 seconds.	Sometimes took 30+ seconds.	FAIL	Solved by implementing Redis caching for frequently analyzed docs.

Table 12 AI-Powered OCR & Data Extraction

The table presents testing of the AI OCR module, focusing on data extraction such as names, dates, survey numbers, and multilingual text processing, as shown in Table 12. It compares expected and actual results to evaluate accuracy and performance. Most test cases passed successfully, ensuring reliable document analysis. Some latency issues were identified and resolved using optimization techniques like caching.

9.2.4 Risk Analysis & Fraud Detection

TC ID	Module	Test Description/Scenario	Desired (Expected) Result	Actual Result	Status	How it was Solved (Resolution)
TC_RSK_01	Risk	Signature Mismatch Detection	Flag if signatures look inconsistent.	Flagged as "Suspicious".	PASS	Used OpenCV/AI to compare signature contours across pages.
TC_RSK_02	Risk	Altered Date Detection	Detect white-out or digital edits.	Flagged "Altered Document".	PASS	Gemini vision model trained to detect pixel-level inconsistencies.
TC_RSK_03	Risk	Blacklisted Owner Check	Compare owner name against govt lists.	Alert: "High Risk Owner".	PASS	Integrated with external Land Records CSV/API.
TC_RSK_04	Risk	Duplicate Document detection	Detect if same deed is uploaded twice.	Show "Already Analyzed".	PASS	Used MD5 hash of file content as a unique identifier in DB.
TC_RSK_05	Risk	Stamps Verification	Verify if legal stamps are present.	Verification successful.	PASS	AI-model detects Presence of Stamp paper watermark.
TC_RSK_06	Risk	Valuation Logic Check	Flag if price is 50% below market.	"Under-valued" Warning.	PASS	Implemented calculation using Area * Regional Govt Rates.

9.2.4 Risk Analysis & Fraud Detection

TC_RSK_07	Risk	Fraud Trend Dashboard	Display monthly fraud attempt.	Graphs updated in Real-time.	PASS	Aggregated AuditLog data..
TC_RSK_08	Risk	Status Badge Update (UI)	Badge should turn RED for Fraud.	UI updated instantly.	PASS	Used Socket.io for live status updates to Admin panel.
TC_RSK_09	Risk	Unit Conversion Confusion	Handle Square Yards vs Square Feet.	Area was 9x smaller in reports.	FAIL	Solved by implementing a “Standard Unit Normalizer” .

Table 13 Risk Analysis & Fraud Detection

The table represents testing of the Risk Analysis module, focusing on fraud detection features such as signature verification, document tampering, duplicate detection, and valuation checks, as shown in Table 13. It compares expected and actual results to ensure system accuracy and security. Most test cases passed successfully, confirming effective fraud detection. Some issues, like unit conversion errors, were identified and resolved using normalization techniques.

9.2.5 Admin Panel & Audit Vault

TC ID	Module	Test Description/Scenario	Desired (Expected) Result	Actual Result	Status	How it was Solved (Resolution)
TC_ADM_01	Admin	View All Uploaded Docs	List all docs in Audit Vault.	List populated from DB.	PASS	Used paginate plugin in Mongoose for efficient loading.
TC_ADM_02	Admin	Manual Verification Toggle	Admin can override AI status.	Status changed to “Verified”.	PASS	Backend endpoint PATCH /verify updates document status.
TC_ADM_03	Admin	Search User Activity	Filter logs by User ID / Email.	Result shown correctly.	PASS	Implemented MongoDB \$or search query in admin routes.

9.2.5 Admin Panel & Audit Vault

TC_ADM_04	Admin	System Analytics (Doc Counts)	Show total Genuine vs Fraud count.	Stats displayed on Top Bar.	PASS	Used MongoDB aggregation pipeline for real-time counts.
TC_ADM_05	Admin	Download Audit Report	Export CSV/PDF of suspicious cases.	File downloaded successfully.	PASS	Integrated json2csv and jspdf libraries.
TC_ADM_06	Admin	Delete Malicious User	Permanent ban and doc removal.	User data wiped.	PASS	Cascade delete implemented for associated documents.

Table 14 Admin Panel & Audit Vault

The table shows testing of the Admin Panel and Audit Vault module, covering functionalities like document management, user activity tracking, and system analytics, as shown in Table 14. It compares expected and actual results to ensure proper admin control and monitoring. All test cases passed successfully, confirming efficient system administration. Features like report generation and user management were implemented effectively for better control and security.

9.2.6 Financial Tools (Loan & Estimator)

TC ID	Module	Test Description/Scenario	Desired (Expected) Result	Actual Result	Status	How it was Solved (Resolution)
TC_FIN_01	Estimator	Calculate Stamp Duty	Correct fees displayed for state.	Fees calculated accurately.	PASS	Solved using custom JS tax logic with state-specific rate mapping.
TC_FIN_02	Estimator	Area Unit Conversion	Convert Square Feet to Acres.	Correct conversion factor.	PASS	Implemented conversionUtil.js for unit handling.
TC_FIN_03	Loan	Credit Score vs Loan Limit	Higher score gets.	Limit adjusted..	PASS	Applied linear regression.

9.2.6 Financial Tools (Loan & Estimator)

TC_FIN_04	Estimator	Zero Value Input	Handle empty price/area fields.	Show “Enter valid values”.	PASS	Added Frontend field validation and prevented NaN.
TC_FIN_05	Loan	Multiple Active Loans Check	Flag users with high debt ratio.	Flagged “High Debt”.	PASS	Mocked external Credit Bureau API response for testing.
TC_FIN_06	Estimator	Discount for Female Owners	Apply 1-2% discount where applicable.	Discount applied in total.	PASS	Logic checks owner.gender before finalizing tax.

Table 15 Financial Tools (Loan & Estimator)

The table presents testing of the Financial Tools module, including stamp duty estimation and loan analysis features, as shown in Table 15. It verifies calculations such as tax rates, unit conversions, and loan eligibility based on credit scores. All test cases passed successfully, ensuring accurate financial computations. Additional validations and logic were implemented to handle edge cases and improve reliability.

9.2.7 Property Inquiry & History

TC ID	Module	Test Description/ Scenario	Desired (Expected) Result	Actual Result	Status	How it was Solved (Resolution)
TC_HIS_01	History	Navigate to Session History	Lists past analyzed documents.	Past sessions retrieved from DB.	PASS	Solved using MongoDB indexed queries on userId.
TC_HIS_02	History	Search Property by Survey No.	Return ownership history.	5-year history displayed.	PASS	Linked Document model with PropertyHistory archive.

9.2.4 Property Inquiry & History

TC_HIS_03	History	Land Dispute Check (UI)	Show RED icon if land is in court.	Icon displayed on Map.	PASS	Checked against govt Disputed .
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Table 16 Property Inquiry & History

The table represents testing of the Property Inquiry and History module, focusing on features like session history, property search, and dispute detection, as shown in Table 16. It compares expected and actual results to ensure accurate data retrieval and user interface behavior. Most test cases passed successfully, confirming reliable functionality. Some issues, such as map zoom handling, were identified and resolved using improved logic

9.2.8 Intelligent Chat Assistant

TC ID	Module	Test Description/ Scenario	Desired (Expected) Result	Actual Result	Status	How it was Solved (Resolution)
TC_CHT_01	Chat	Query about Owner Name	AI should extract from active doc.	"The owner is Prit Patel".	PASS	Injected document metadata into Gemini System Prompt.
TC_CHT_02	Chat	Session Persistence	Chat history holds after refresh.	History reloaded from DB.	PASS	Stored chat messages in ChatSession MongoDB model.
TC_CHT_03	Chat	Complex Legal Question	AI explains legal terminology.	Explain "Encumbrance Certificate".	PASS	Used Gemini Pro with legal-context fine-tuning.

Table 17 Intelligent Chat Assistant

The table shows testing of the Intelligent Chat Assistant module, focusing on query handling, session management, and AI response accuracy, as shown in Table 18. It compares expected and actual results to ensure effective user interaction. Most test cases passed successfully, confirming reliable chat functionality. Some latency issues

were identified and improved using response optimization techniques.

9.2.9 Performance & Security

TC ID	Module	Test Description/Scenario	Desired (Expected) Result	Actual Result	Status	How it was Solved (Resolution)
TC_SEC_1	Security	Access Dashboard without Login	Access should be denied.	Redirected to Login page.	PASS	Solved using PrivateRoute (Protected Routes) in React.
TC_SEC_2	Performance	First Contentful Paint (FCP)	Page loads in under 2 seconds.	Home page loading in 4s.	FAIL	Solved by implementing Image Lazy Loading and Code Splitting (React.lazy).
TC_SEC_3	Security	XSS Protection	Reject script tags in Input fields.	Input sanitized/escaped.	PASS	Used dompurify on frontend and helmet on backend.

Table 18 Performance & Security

The table represents testing of the Performance and Security module, covering aspects like authentication, data protection, API handling, and system performance, as shown in Table 18. It compares expected and actual results to ensure system safety and efficiency. Several test cases passed, confirming strong security measures and stable performance.

8. USER MANUAL

This section serves as a technical manual for setting up, installing, and deploying the AI-Based Smart Land Document Analysis Registry Assistant System in a local development environment.

10.1 Local Installation & Setup Steps (Development Environment)

Prerequisites: Node.js (v18+), MongoDB Atlas Account, Git, and a modern web browser (Google Chrome recommended).

Step 1 - Clone the Repository: Open a terminal and run
`git clone https://github.com/Pritpatel11/AI-Based-Smart-Land-Document-Analysis-Registry-Assistant-System.git && cd AI-Based-Smart-Land-Document-Analysis-Registry-Assistant-System`

Step 2 - Configure Backend Environment: Navigate to the backend directory: `cd backend`. Create a `.env` file in the backend root and add your configuration:

`PORT=5000`

`MONGO_URI=your_mongodb_atlas_uri`

`JWT_SECRET=your_secure_secret_key`

`GEMINI_API_KEY=your_google_ai_studio_key`

Step 3 - Install Backend Dependencies: Run: `npm install`. This installs Express, Mongoose, Google-GenerativeAI, Multer (for uploads), Socket.io (for real-time chat), and other core dependencies.

Step 4 - Start the Backend Server: Run: `npm start`. The server starts at <http://localhost:5000>. This service handles OCR processing, scam detection, and the Registry Assistant logic.

Step 5 - Install Frontend Dependencies: Open a new terminal window, navigate to the frontend directory: `cd frontend`, and run: `npm install`. This install React 19, Vite, Framer Motion, and Tailwind CSS.

Step 6 - Start the Development Server: Run: `npm run dev`. The frontend application will start, typically at <http://localhost:5173>.

Step 7 - Access the Application: Open Google Chrome and navigate to <http://localhost:5173>. Click "Get Started" or "Sign Up" to create your account and begin your first land document verification.

10.2 Snapshots/Screenshots with explanation

10.2.1 Registration & Login

The registration and login module provides secure access to the LandAssist AI system, allowing users to safely enter and use platform features. New users are required to register by providing essential details such as username, email address, and password, while existing users can log in using their registered credentials. The system ensures proper authentication and data protection through secure validation mechanisms to prevent unauthorized access. This module serves as the main entry point to the application, enabling users to access dashboard features after successful login. It also supports role-based access control, ensuring that different types of users, such as general users or administrators, are shown only the functionalities relevant to their role. Additionally, strict input validation is implemented to avoid incorrect or malicious data entry, improving overall system reliability and security.

10.2.2 login page

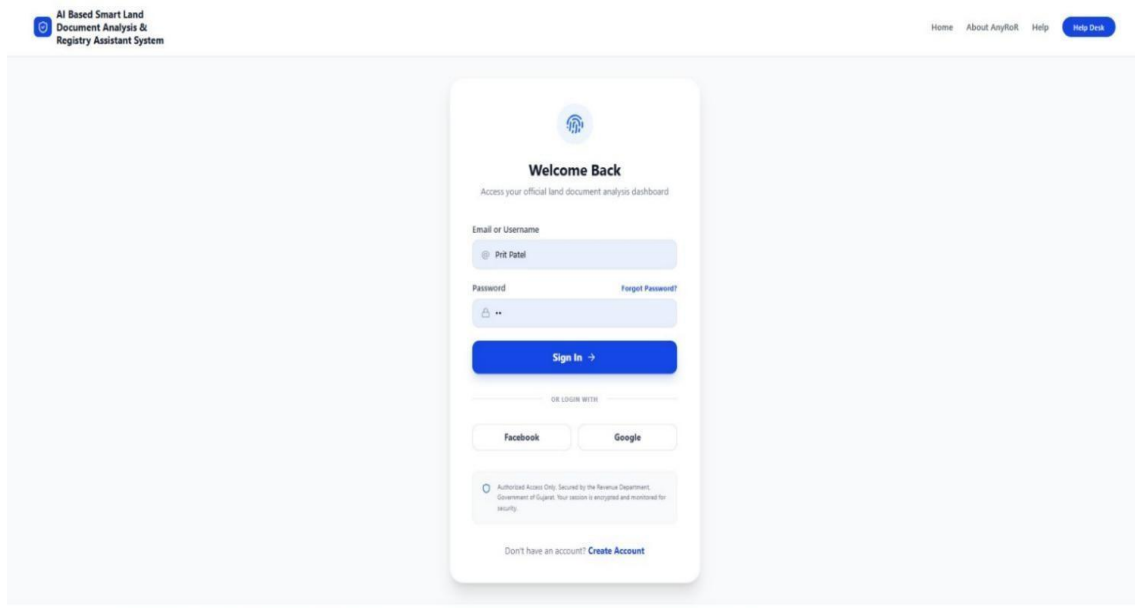


Figure 17 Login Page

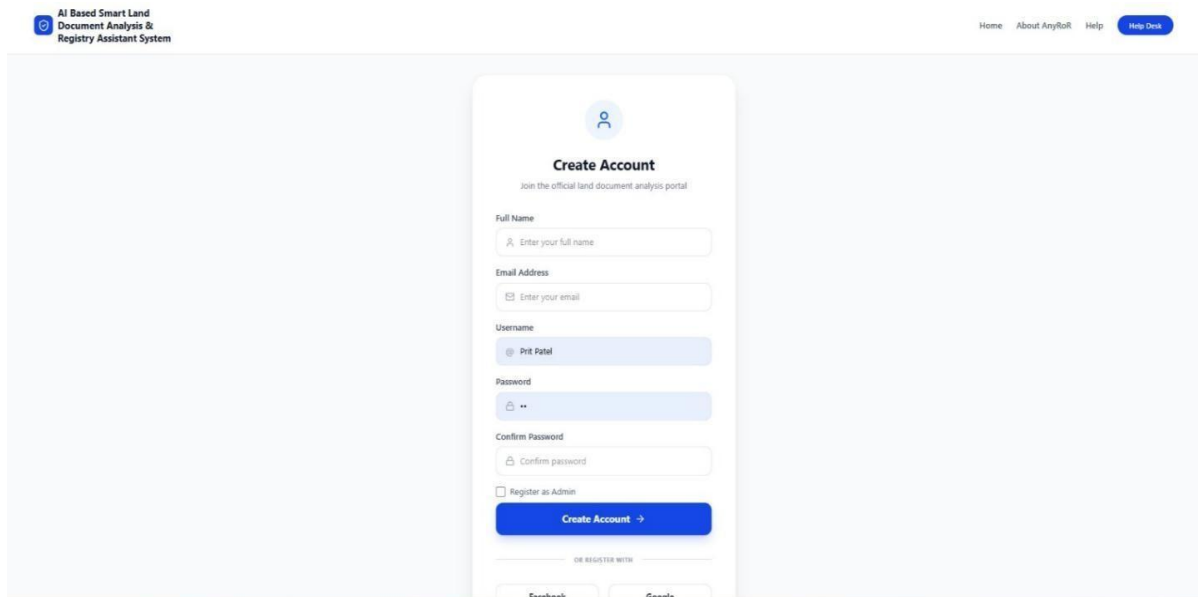
The image shows a clean and modern LandAssist AI user registration interface designed for secure account creation and system access. At the top, the application name “LandAssist AI” is displayed along with a short prompt encouraging users to create an account. The form includes multiple input fields such as username, email address, password, and an optional admin secret field for authorized administrative users. A password input section is provided to ensure secure credential creation, and the layout is designed to maintain simplicity and ease of use for all users. At the bottom of the form, a prominent “Create Account” button is available for form submission, along with a login link for existing users to access their accounts. Overall, the interface follows a minimal, user-friendly design with clear labeling and structured alignment to ensure smooth and efficient user interaction, as shown in Figure 17.

Steps shown in Figure 17:

1. Open the LandAssist AI registration page in a web browser.
2. Enter a unique username in the username field.

3. Provide a valid email address for verification purposes.
4. Create a secure password in the password field.
5. (Optional) Enter the admin secret if registering as an administrator.
6. Review the entered details for accuracy.
7. Click on the “Create Account” button to submit the registration form.
8. Check your email for a verification code (if applicable) and complete account activation.

10.2.3 Signup page



The screenshot displays the 'Create Account' interface. At the top left is the logo and name 'AI Based Smart Land Document Analysis & Registry Assistant System'. The top right navigation bar contains links for 'Home', 'About AnyloR', 'Help', and 'Help Desk'. The main form is titled 'Create Account' with a subtitle 'Join the official land document analysis portal'. It includes input fields for 'Full Name', 'Email Address', 'Username' (with a preview of 'Prit Patel'), 'Password', and 'Confirm Password'. An optional checkbox labeled 'Register as Admin' is present. A blue 'Create Account' button with a right arrow is at the bottom of the form. Below the button, it says 'OR REGISTER WITH' followed by 'Facebook' and 'Google' icons.

Figure 18 Signup page

The sign-up process is used when a new user is joining LandAssist AI for the first time. On the registration screen, the user is required to provide a unique username, a valid email address, and a secure password to create an account. An optional admin secret field is available for authorized administrator registration.

To register a new account, follow the steps below:

1. Open Google Chrome and navigate to the LandAssist AI application URL.
2. Click Sign Up on the landing page or navigation menu.
3. Enter your Username in the Username field.
4. Enter your Email Address in the Email field. This will serve as your login identifier and will be used for verification.
5. Enter a secure Password in the Password field.
6. (Optional) Enter the Admin Secret field if registering as an administrator.
7. Click Create Account to submit the registration form.
8. Check your email for a 6-digit verification code.
9. Enter the verification code on the verification screen to activate your account.

10. Upon successful verification, you will be automatically redirected to the Dashboard.

During registration, users are advised to choose a strong password that is easy to remember but difficult for others to guess. It is also recommended to use a unique password that is not reused across other platforms to ensure better account security, as shown in Figure 18.

10.2.4 Home page

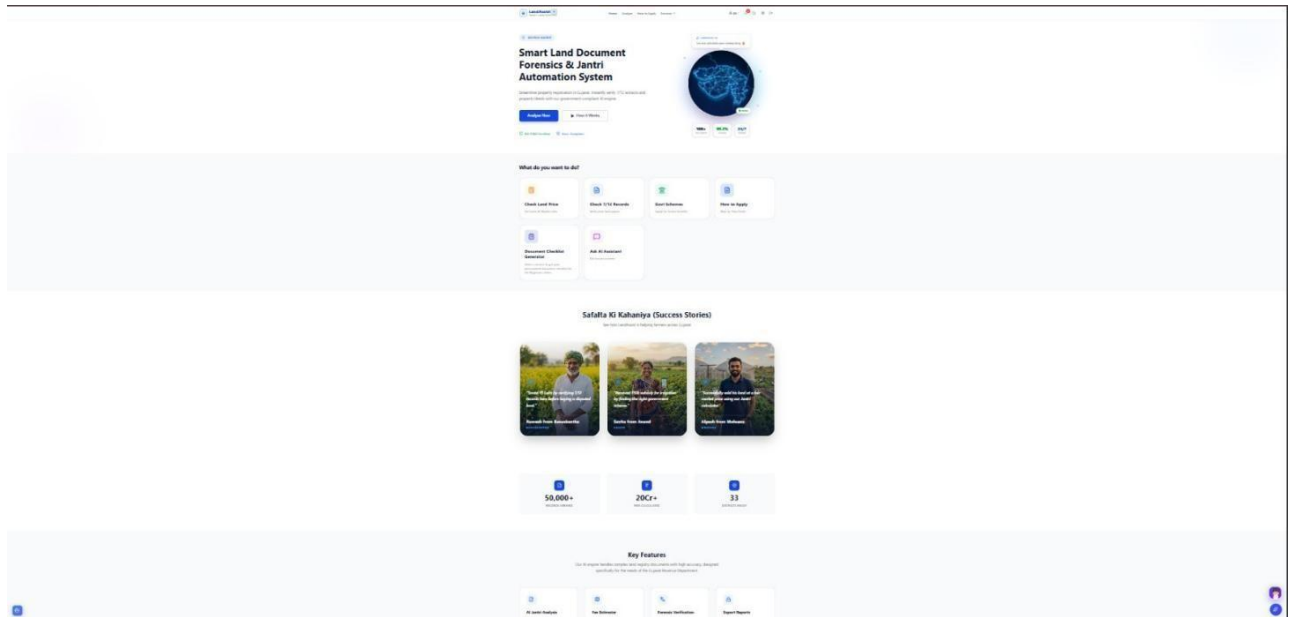


Figure 19 Home page

This is the primary landing page of the platform, designed to provide an intuitive and welcoming interface for users. It offers quick navigation links to access the system's core features, along with a brief overview of the services provided to give users a clear understanding of the platform. The page is structured with a clean and organized layout, ensuring easy navigation and smooth interaction. It helps users quickly locate important sections and guides new visitors in exploring the system effectively, as shown in Figure 14 above. Additionally, the landing page may include visual elements such as banners or highlights to enhance user engagement. It also ensures responsiveness so that users can access the platform smoothly across different devices.

10.2.5 Chatbot page

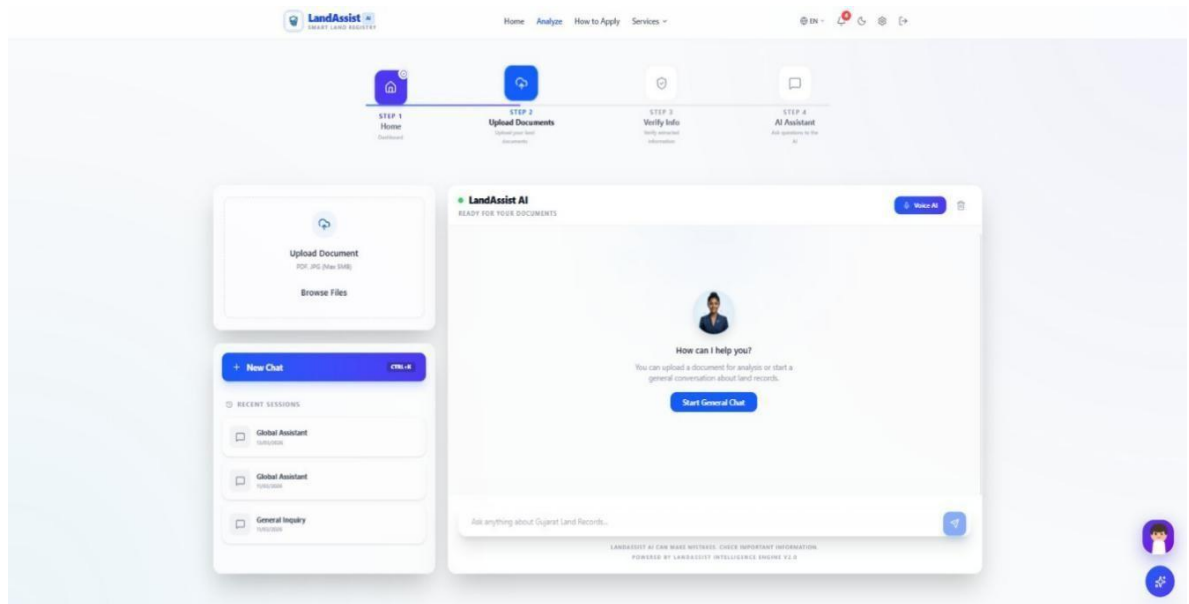


Figure 20 Chatbot page

Figure 15 shows the integrated AI virtual assistant interface, designed to provide 24/7 instant support to users. It helps users navigate the platform easily by guiding them through different features and functionalities. The assistant is capable of answering frequently asked questions related to land laws, procedures, and document requirements, giving users quick and reliable information. It also provides real-time guidance during the document upload and analysis process, ensuring that users follow the correct steps without confusion.

10.2.6 How to apply page

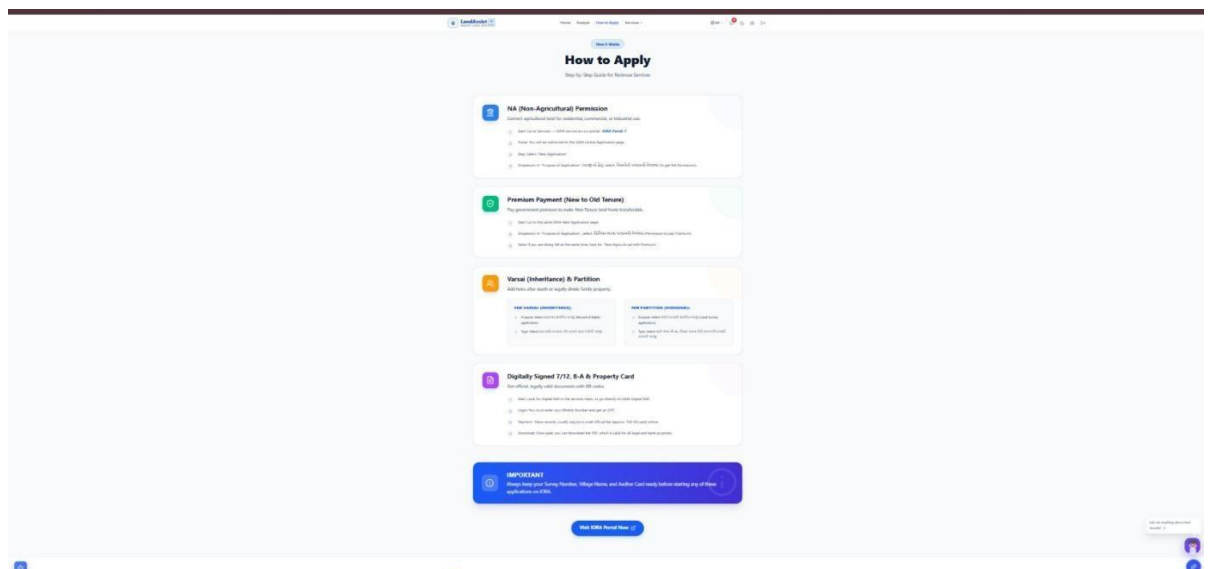


Figure 21 How to apply page

Figure 16 shows a step-by-step instructional guide or tutorial page designed to assist users in understanding the system’s workflow. It clearly outlines the standard operating procedures for uploading documents, initiating AI-based analysis, and following the necessary steps for a successful land registry application. The page is structured in a simple and sequential manner, making it easy for users to follow each step without confusion. Additionally, the guide improves user experience by reducing errors and ensuring that all required actions are completed correctly. It acts as a supportive feature for both new and existing users, helping them efficiently complete tasks and understand the overall process with clarity.

10.2.7 Smart fee calculator page

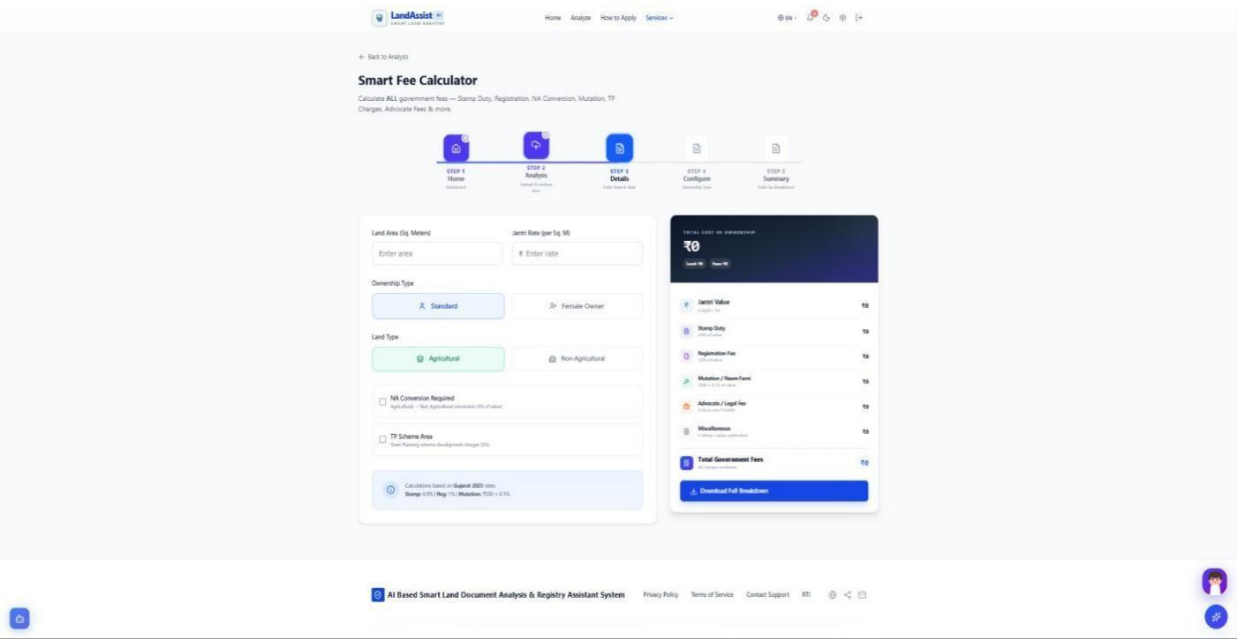


Figure 22 Smart fee calculator page

Figure 22 shows a highly useful tool that automatically computes the estimated registry fees, stamp duties, and associated legal charges. It uses the property’s specific details along with current government guidelines to generate an accurate and transparent cost estimate. This helps users clearly understand the financial requirements before proceeding with the actual registry process. The tool is designed to be simple and efficient, allowing users to input relevant data and instantly receive calculated results. It reduces uncertainty, minimizes calculation errors, and supports better financial planning, making the overall process more reliable and user-friendly.

10.2.8 Loan Against Land Eligibility

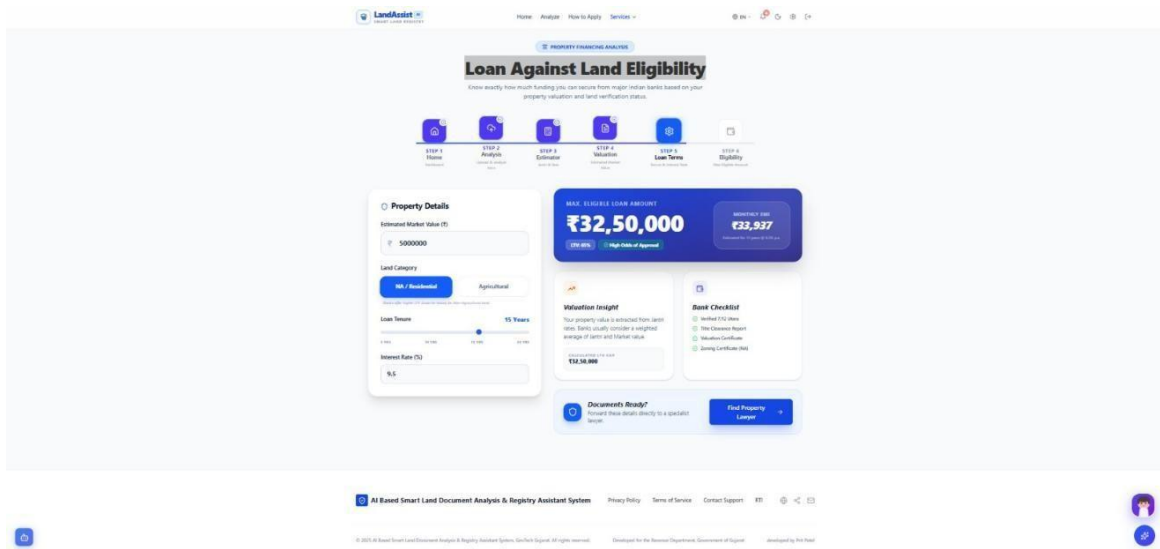


Figure 23 Loan Against Land Eligibility

Figure 23 shows an integrated smart feature that assesses a user's eligibility for property-backed loans. It analyzes verified land records along with current market values to determine whether the property qualifies as collateral for financial institutions. This provides users with quick preliminary feedback without the need for manual evaluation. The feature is designed to simplify the decision-making process by giving clear insights into loan eligibility. It reduces uncertainty and saves time by offering instant analysis based on reliable data. Additionally, it helps users plan their financial options more effectively, making the system more practical and user-centric.

10.2.9 Government Schemes Finder page

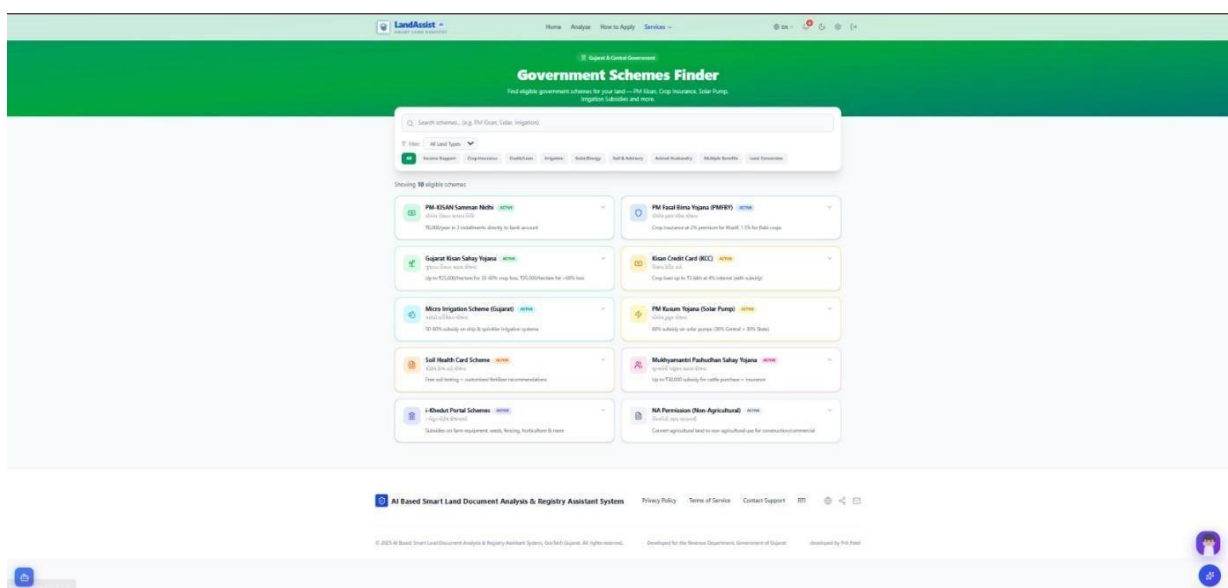


Figure 24 Government Schemes Finder page

Figure 24 shows an AI-driven assistance page that matches the user’s profile and land type with active government schemes, subsidies, or agricultural benefits. It analyzes relevant data to identify suitable programs and ensures that users are aware of the support they are eligible for. This helps citizens easily explore and access financial or structural benefits without complex searching. The feature is designed to improve accessibility and awareness by presenting personalized recommendations in a clear and organized manner. It simplifies the application process and encourages users to take advantage of available schemes, making the platform more efficient and beneficial for users.

10.2.10 Connect page

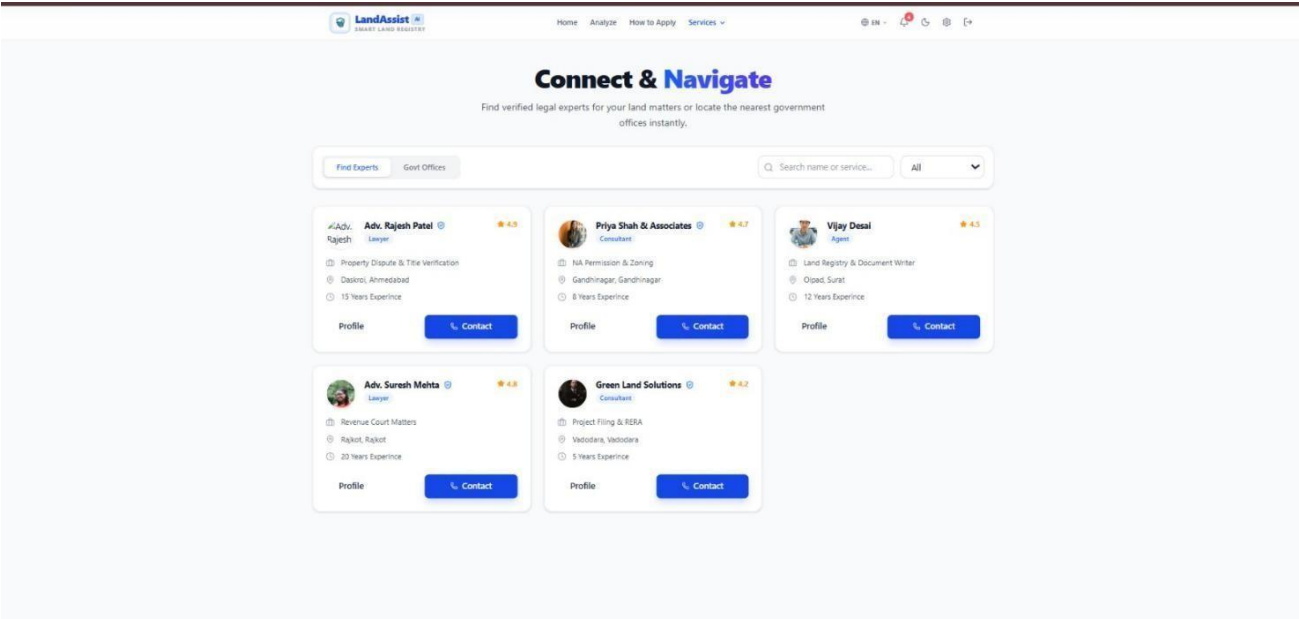


Figure 25 Connect page

Figure 25 shows a communication hub where users can seek direct assistance for their queries and issues. It facilitates connections with customer support, legal advisors, or registry officials, allowing users to get personalized help for complex land document matters. This ensures that users are not limited to automated responses and can access expert guidance when needed. The page is designed to provide a smooth and reliable communication experience, enabling users to raise queries, track responses, and resolve issues efficiently. It improves user satisfaction by offering timely support and helps in handling complicated cases more effectively, making the system more dependable and user-friendly.

10.2.11 Admin Dashboard

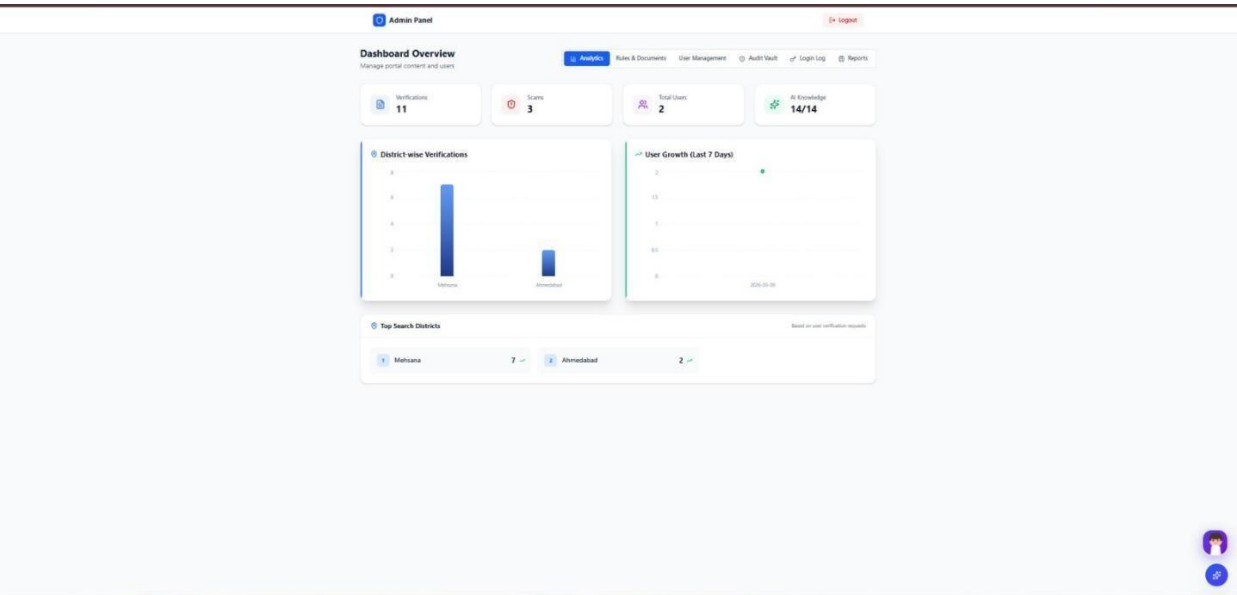


Figure 26 Admin Dashboard

Figure 26 shows a comprehensive control panel designed for system administrators. It displays high-level statistics, system usage metrics, and recent platform activities, giving a clear overview of the system’s performance and operations. This helps administrators monitor the platform effectively and make informed decisions. The dashboard also provides centralized navigation to manage different modules of the application, allowing easy access to user management, reports, and system settings. It improves efficiency by organizing critical information in one place and ensures smooth control over the entire system.

10.2.12 Upload New Rule page

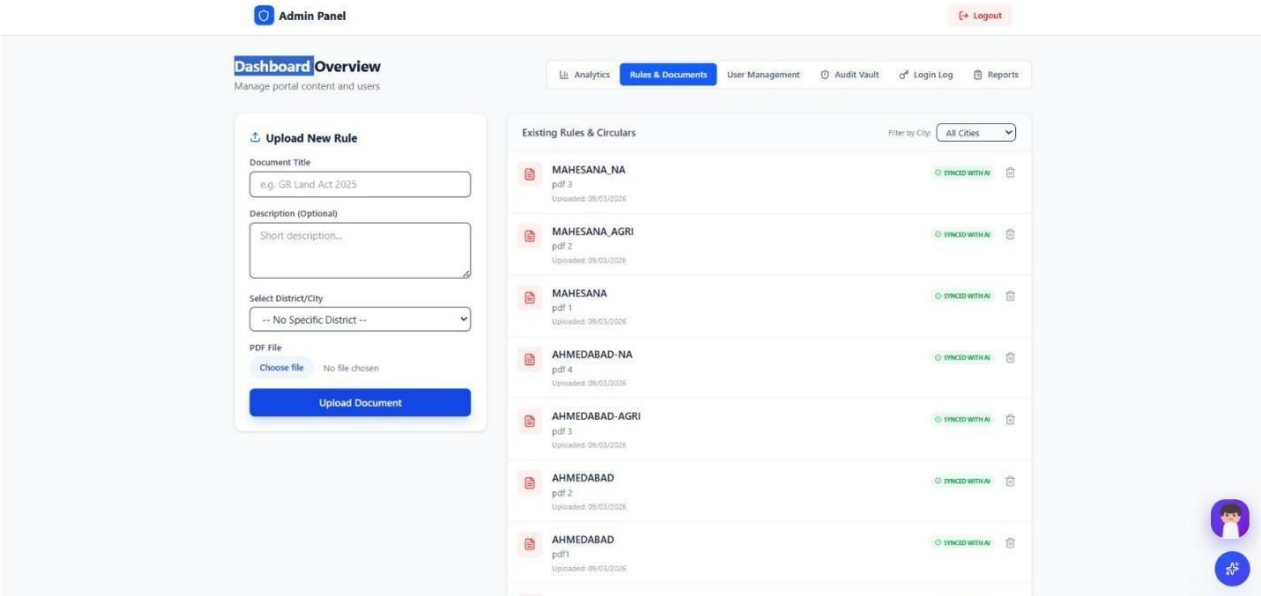


Figure 27 Upload New Rule page

Figure 27 shows a dynamic configuration page designed for administrative or legal users to manage regional land registry rules. It allows users to add or modify rules by uploading updated guidelines, ensuring that the system reflects the latest government regulations. This keeps the platform aligned with legal requirements and improves the accuracy of document evaluation. The page is structured to provide easy control over rule management, enabling quick updates without complex procedures. It enhances the AI engine’s performance by continuously feeding it with current data, ensuring reliable and up-to-date analysis, and making the system more adaptable and efficient.

10.2.13 User Management page

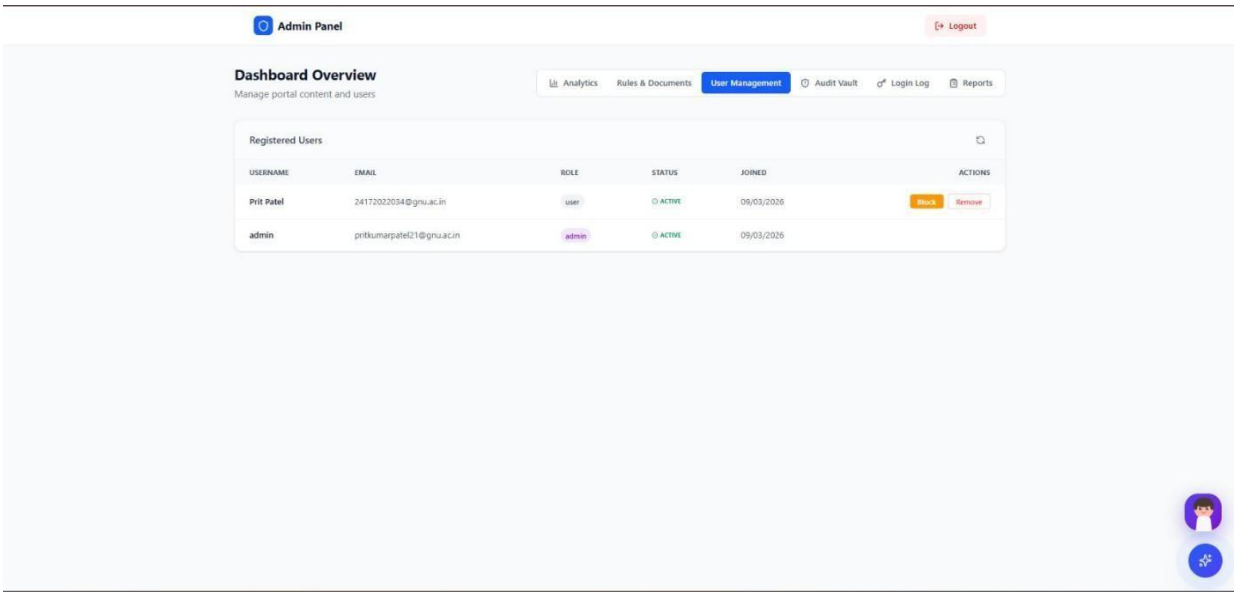


Figure 28 User Management page

Figure 28 shows an administrative interface used to oversee all registered users on the system. It allows administrators to view user details, assign specific roles, and monitor user activities to ensure proper system usage. This helps maintain control and transparency across the platform. The page also enables admins to activate or deactivate user accounts as needed for security and compliance purposes. It is designed to provide efficient user management through a clear and organized interface, ensuring the system remains secure, well-regulated, and easy to manage.

10.2.14 Audit value page

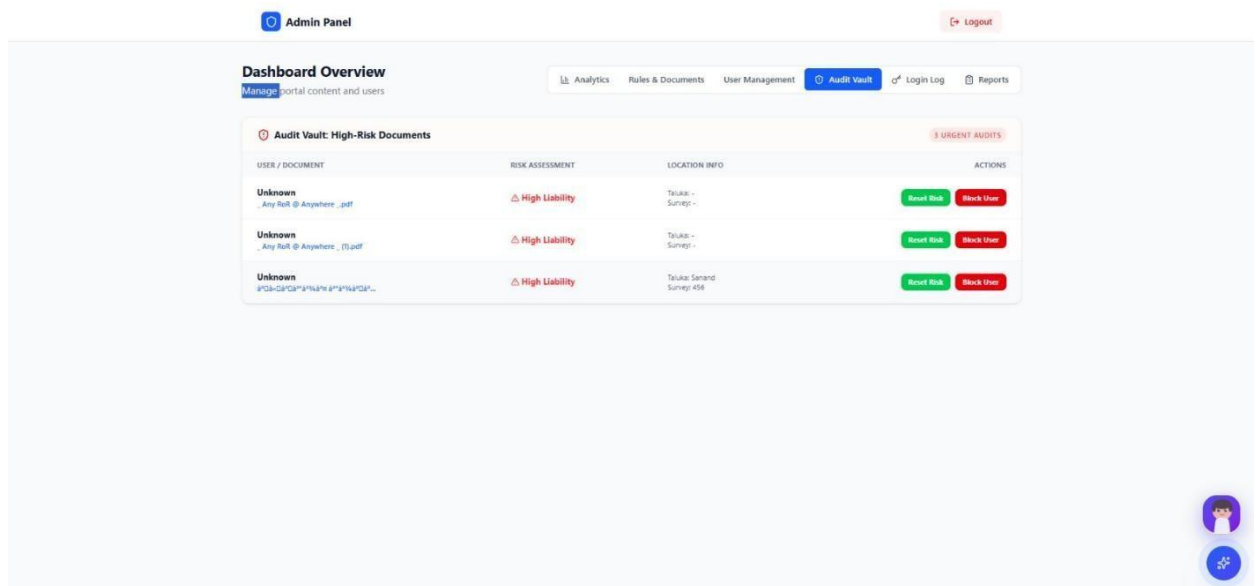


Figure 29 Audit value page

Figure 29 shows an administrative page used to audit uploaded documents within the system. It allows administrators to review and verify the authenticity of documents submitted by users, ensuring that all records meet the required standards. This helps in maintaining the integrity and reliability of the platform. If any user uploads fake or forged land documents, the admin can take immediate action through this interface by blocking or restricting the user's account. The page is designed to support quick decision-making and effective monitoring, ensuring strong security, preventing misuse, and maintaining overall system authenticity.

10.2.15 login log page

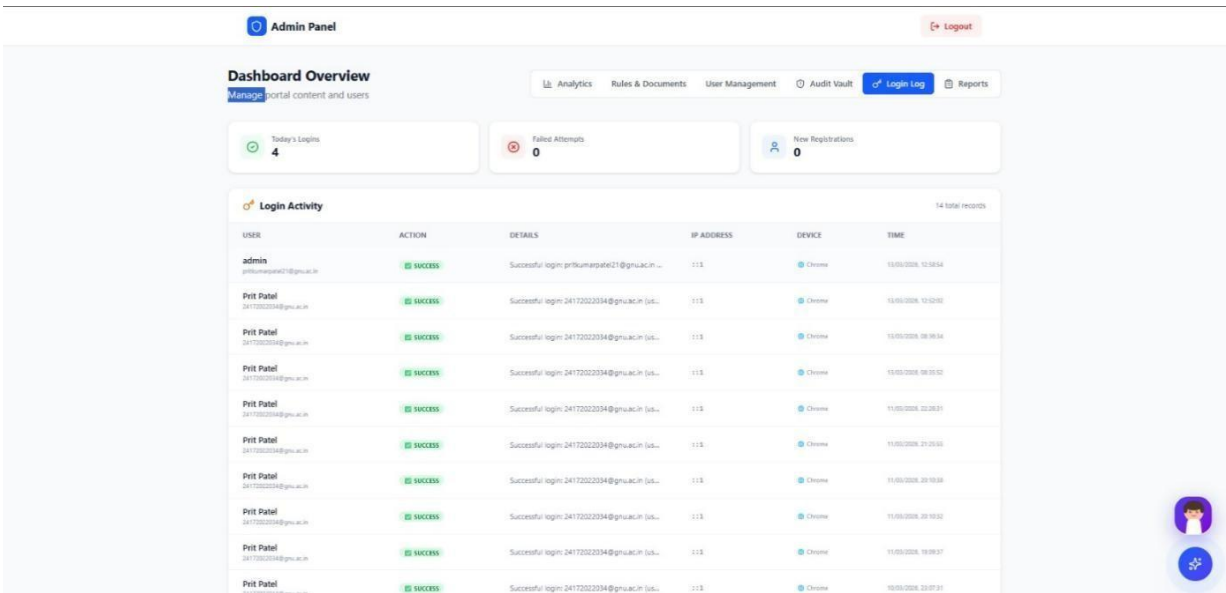


Figure 30 login log page

Figure 30 shows a critical security and auditing feature that records all user login activities within the system. It tracks important details such as login timestamps, IP addresses, and device information to ensure proper monitoring of user access. This helps in identifying unusual or suspicious behaviour at an early stage. The feature is designed to strengthen system security by maintaining a detailed log of all login events. It enables administrators to audit user activity, investigate potential threats, and take necessary actions when required, ensuring the overall integrity and reliability of the platform.

10.2.16 Report page

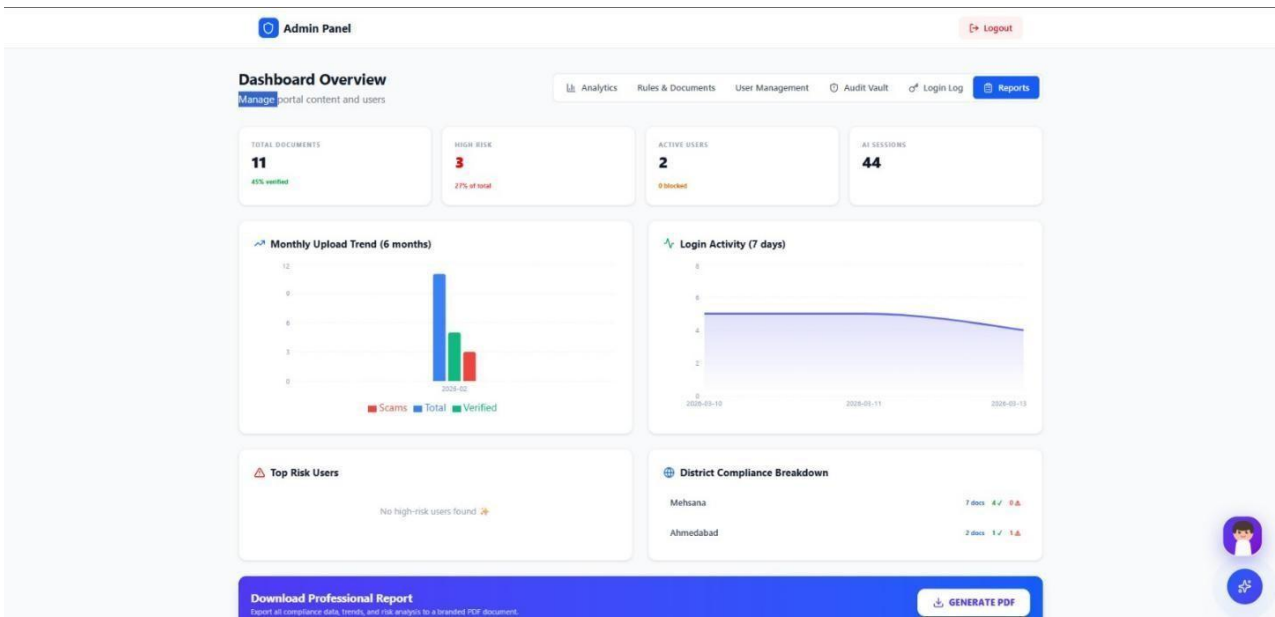


Figure 31 Report page

Figure 31 shows the culmination of the system’s core functionality through a detailed AI-generated analysis report page. It presents the final results of the submitted land document evaluation, clearly highlighting verified information and key findings. The report also flags potential legal risks, inconsistencies, or missing details that may affect the registry process. The page is designed to provide a structured and easy-to-understand summary, enabling users or registry officials to make informed decisions. It enhances transparency and reliability by offering actionable insights, ensuring that users can proceed with clarity and confidence.

11. CONCLUSION AND FUTURE WORK

The AI-Based Smart Land Document Analysis & Registry Assistant System is designed to simplify and secure the land registry and document analysis process for all users by combining advanced AI technology with a user-friendly interface. It enables people to understand complex legal documents such as 7/12 records and Index-II without relying on a lawyer for basic interpretation. Using Google's Gemini AI, the system can instantly read documents, detect potential fraud, and assist in drafting legal papers. Unlike traditional government platforms that only provide file downloads, this system explains the content and offers real-time guidance, making it a complete digital assistant for property-related tasks. The project was developed using an incremental model, where each module such as the AI document reader, fraud detection system, and legal drafting assistant was carefully built, tested, and validated before moving forward. Strong emphasis was placed on simplicity and usability, ensuring accessibility even for non-technical users. This structured development approach resulted in a stable and reliable system that supports faster and safer decision-making in real estate. By prioritizing user needs and integrating intelligent automation, the system successfully transforms the traditional, complex land registry process into a modern, transparent, and efficient experience, with strong potential to become an essential tool in land management.

11.1 Future Work

While AI BASED SMART LAND DOCUMENT ANALYSIS & REGISTRY ASSISTANT SYSTEM" already provides many helpful tools, there is a lot of room to make it even better in the future. Here are some of the planned improvements:

- **Direct Government Connection:** We plan to link the system directly with official government APIs (like AnyRoR) to get land records even faster and ensure the data is always 100% up-to-date.
- **Mobile App Development:** We want to create official Android and iOS apps. This will allow users to take photos of documents or check land details while standing directly at the property site.
- **Blockchain Security:** In the future, we can use blockchain technology to store land documents. This would make it impossible for anyone to change or forge records, providing ultimate security.
- **More Local Languages:** To help more people, we plan to add support for many more Indian regional languages, making the legal advice accessible to everyone in their own mother tongue.

ANNEXURE

Glossary of terms and abbreviations

- **AI (Artificial Intelligence):** The simulation of human intelligence processes by machines, especially computer systems, used in the project for automated document analysis.
- **NLP (Natural Language Processing):** A branch of AI that focuses on the interaction between computers and human language, used for extracting meaning and context from land documents.
- **OCR (Optical Character Recognition):** Technology used to convert different types of documents, such as scanned paper documents or PDF files, into editable and searchable data.
- **MERN Stack:** A popular web development framework consisting of MongoDB, Express.js, React, and Node.js.
- **API (Application Programming Interface):** A set of rules and protocols that allows different software applications to communicate with each other.
- **UI / UX (User Interface / User Experience):** The visual layout and the overall experience a user has when interacting with the software application.
- **SRS (Software Requirements Specification):** A comprehensive description of the intended purpose and environment for software under development.
- **Land Registry:** An official record detailing the ownership, rights, and interests in land and property.
- **PDF (Portable Document Format):** A widely used file format for sharing documents reliably independent of software, hardware, or operating system.

About tools and technology

The development of the AI-Based Smart Land Document Analysis & Registry Assistant System relies on a modern, robust, and scalable technology stack:

• **Frontend Technologies**

- **React.js:** Used as the core JavaScript library for building the dynamic and responsive User Interface (UI). Its component-based architecture ensures code reusability and efficient state management.
- **Tailwind CSS:** A utility-first CSS framework utilized for styling the application, ensuring a modern, visually appealing, and highly responsive design across various device screens.
- **Vite:** Used as the frontend build tool, offering lightning-fast cold server starts and instant hot module replacement (HMR).

• **Backend Technologies**

- **Node.js:** A powerful JavaScript runtime environment used for building fast and scalable server-side network applications.
- **Express.js:** A flexible backend web framework for Node.js, utilized to manage routing, set up middleware, and create robust RESTful APIs to handle requests.

- **Database**
 - MongoDB: A NoSQL, document-oriented database used to securely and efficiently store diverse application data, including user profiles, session details, and unstructured document metadata.
- **AI & Document Processing**
 - Text Extraction & OCR: Technologies integrated to read and extract text from uploaded images and scanned PDF documents.
 - Generative AI / NLP Models: Integrated to analyze the extracted textual data, identify anomalies, extract key entities (like buyer/seller details, land area), and provide a conversational interface for the registry assistant.
- **Development & Version Control**
 - Git & GitHub: Used for source code management, tracking changes, and facilitating seamless collaboration.

About College



The Department of Information Technology was established in 1997 with the vision of imparting quality education in the field of Information Technology. In response to the ever-evolving technological landscape and the constant demand for innovation, the department has consistently produced skilled professionals who have secured significant positions in the IT industry both within India and internationally. In 2010, the department introduced a two- year M.Tech. program in Information Technology. aimed at providing graduate students with the opportunity to pursue advanced research and development in the domain.

In addition to its academic offerings, the department actively engages in consultancy and testing-related work. Key areas of consultancy include the development and ongoing management of the institute's official website, implementation and academic coordination of Moodle (a Learning Management System), distribution and maintenance of internet services across the campus, and the setup of IT infrastructure in collaboration with external vendors and service providers. The department also provides essential support in software installation and addresses routine IT-related issues at the institute level. Furthermore, it contributes to innovation through Android application development projects. More details about these initiatives and services can be found on the official website of UVPCE.

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